

JAPAN PORTFOLIO STRATEGY

More than meets the AI: a new framework for Japan's expanding AI universe

Japan's tech-heavy Nikkei 225 index experienced a peak-to-trough correction of -15% from June 25 to July 29, with the NT (Nikkei-to-TOPIX) ratio falling from an all-time-high of 18.0x to 15.4x over the same period. The Nikkei 225 has since rebounded +9%, but sentiment toward AI-linked names, which drove much of its earlier advance during CY2026, is currently being reassessed. That reassessment has been complicated by the fact that the definition of an 'AI-related stock' has widened considerably over the last few years. The AI thematic now extends well beyond processors and semiconductor equipment to encompass optical interconnects, power and cooling infrastructure, robotics, and applications. Faced with this broader opportunity set, we have decided to present a more systematic categorization of Japanese stocks across the AI value chain. We therefore focus on a detailed analysis of the Japanese semiconductor ecosystem (SM), as well as the data center hardware (HW) stocks which underpin the Japan AI thematic. The market cap and 6M advt thresholds for this analysis were set at ¥50bn and US\$5mn respectively. Our key conclusions are:

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- **Japan represents an integral part of the global AI universe:** Japan AI SM/HW stocks account for 25% of TOPIX, and 35% of the corresponding AI SM/HW market cap in Asia. By sector, the Japan AI opportunity set tends to be more concentrated in materials, SPE, rack components, and data center infrastructure.
- **Higher earnings growth and valuations for the market, but with higher volatility:** AI SM/HW related stocks are expected to contribute to one-third of FY26/FY27 TOPIX EPS growth. However, higher levels of position concentration are likely to increase the frequency and severity of AI-driven market corrections.
- **Introducing GS Japan AI Semiconductor Ecosystem basket (GSJPAISM) and the GS Japan AI Data Center Hardware basket (GSJPAIHW):** Within the broader Japan AI SM/HW universe, we focus on stocks with higher liquidity and AI-related exposure and introduce two new baskets to track the performance of the AI SM/HW thematic.
- **Japan's AI universe is too important to ignore:** Given its contribution to index-level earnings growth and price performance, the risks of not being meaningfully exposed to the AI thematic in Japan are likely to persist. However, managing the increased levels of portfolio volatility this creates will be crucial.

AI is reshaping the Japanese equities market

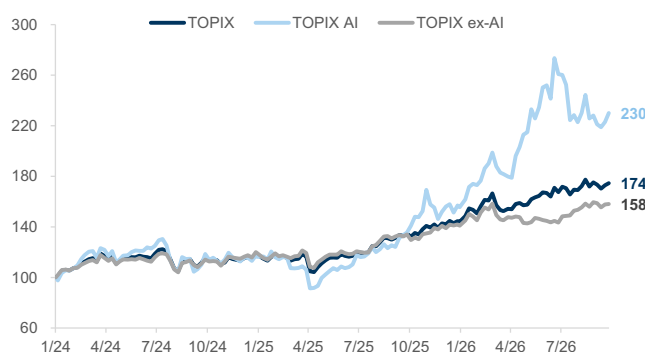
Our main objective with this report is to provide a comprehensive framework for analyzing Japan's rapidly expanding AI universe, and also to position the relative competitiveness of the Japanese AI value chain within the broader global AI ecosystem. The report highlights AI's growing influence on Japanese domestic equity performance and earnings, it categorizes the current Japanese AI universe based on RBICS sub-industry revenue exposures, and then analyzes the differences in growth and valuation profiles across the various Japanese AI value chain sub-layers. We then use this framework to define the **Japan AI Semiconductor Ecosystem universe (SM universe)** and the **Japan AI Data Center Hardware universe (HW universe)**. We also highlight Physical AI as a major medium-term Japanese AI application thematic, and attempt to discuss macro headwinds and tailwinds for the overall AI thematic.

AI-related stocks (here, we refer to the SM and HW universe + SoftBank Group) have been important drivers of Japanese equity performance in recent years. Since the start of 2024, and as of September 25, TOPIX has produced a cumulative return of +74%, compared with +58% for TOPIX excluding AI-related constituents, and +130% for just its AI-related constituents ([Exhibit 1](#)). The performance gap highlights the impressive strength of AI-linked companies relative to the rest of the market.

AI exposure also differs across Japan's two main equity benchmarks, and under our mapping criteria, AI-related stocks account for 48% of the Nikkei 225 and 25% of TOPIX ([Exhibit 2](#)). 76% of Nikkei's CY2026 ytd price appreciation of +37%, and 50% of TOPIX's CY2026 ytd price appreciation of +21% can be attributed to what we have categorized as AI-related stocks.

Exhibit 1: AI-related stocks in TOPIX significantly outperformed TOPIX ex-AI

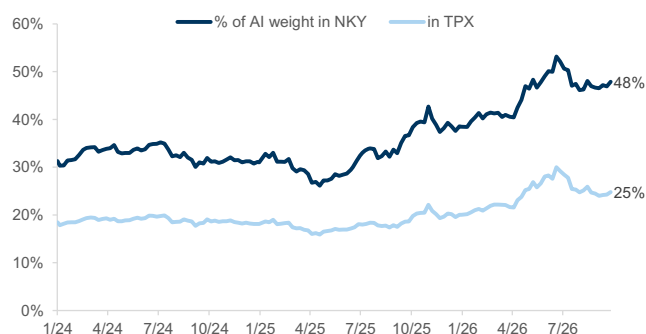
Data as of 25 September 2026



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

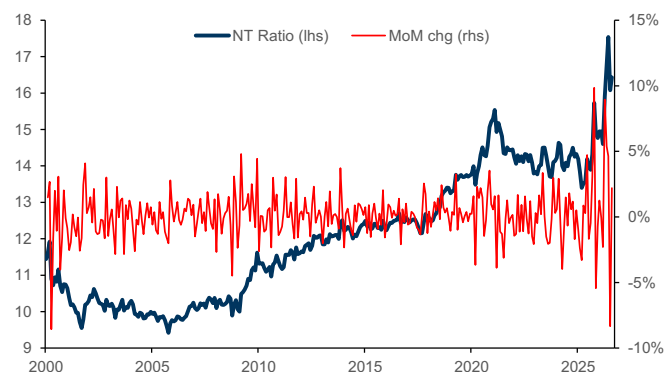
Exhibit 2: Weight of AI-related stocks in Nikkei 225 is close to double of that in TOPIX

Data as of 25 September 2026

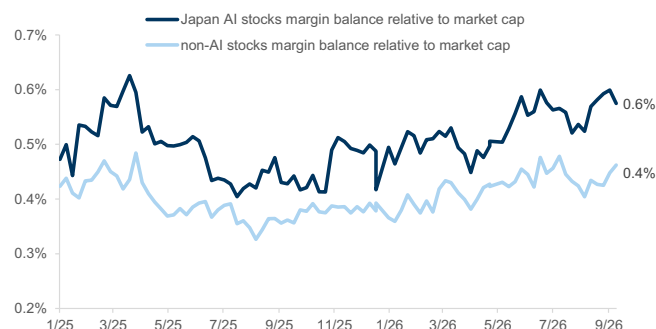


Source: FactSet, Goldman Sachs Global Investment Research

The NT ratio jumped from 14.5x in early April to an all-time-high of 18.0x by the end of June, experiencing some of the most rapid mom moves in its history during this period. We believe that the difference in AI-exposure highlighted above was one of the main drivers of both Nikkei's recent out-performance versus TOPIX, as well as the sudden increase in NT ratio volatility ([Exhibit 3](#)).

Exhibit 3: Outperformance of AI stocks led to a surge in NT ratio and an increase in its volatility


Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

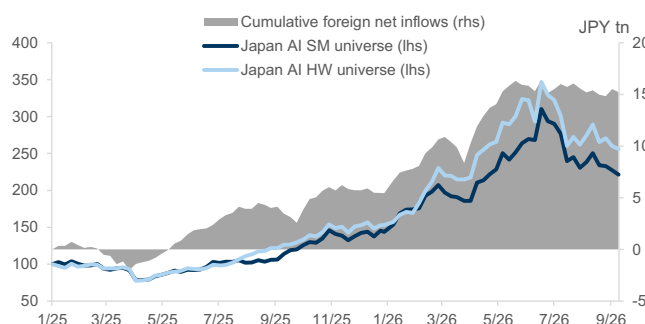
Exhibit 4: AI-related SM/HW stocks have higher margin balance relative to market cap compared to non-AI stocks
 Data as of Sep 18, 2026


Source: JPX, Data compiled by Goldman Sachs Global Investment Research

The strong performance and higher volatility around AI-related names can in part be attributed to a combination of higher margin balances relative to non-AI counterparts ([Exhibit 4](#)), as well as higher foreign investor participation that tends to be more dynamic versus other investor types ([Exhibit 5](#)). As can be seen in [Exhibit 4](#), AI SM/HW-related stocks' margin balance relative to market cap is 1.5x that of non-AI SM/HW stocks.

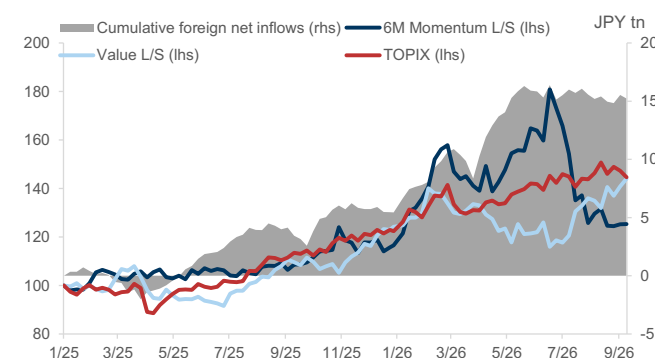
Exhibit 5: AI-related stocks have been driven by strong foreign inflows

Data as of Sep 11, 2026



Semi-annually rebalanced performance based on average of market cap and liquidity

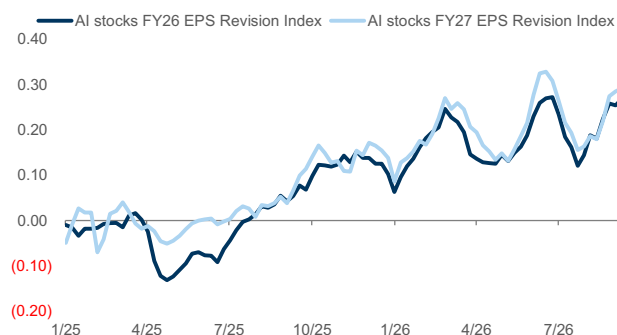
Source: FactSet, JPX, Data compiled by Goldman Sachs Global Investment Research

Exhibit 6: TOPIX has been resilient with rotation into value, but will likely require recovery of AI-stocks to go significantly higher


Source: FactSet, JPX, Data compiled by Goldman Sachs Global Investment Research

Earnings revisions provide an additional fundamental argument for the importance of Japan's AI thematic. Our earnings revision index shows favorable revision trends for AI-related stocks across both FY26 and FY27, indicating that expectations of positive earnings momentum could continue over a multi-year time-frame. As can be seen in [Exhibit 8](#) below, our **Japan AI SM universe** and **Japan AI HW universe** are expected to deliver FY26 yoy EPS growth of +105% and +67% respectively versus +22% for TOPIX. Earnings growth rates for these two universes in FY27 and FY28 are also expected to be well above TOPIX levels.

Exhibit 7: Sell-side analysts continue to revise up AI stocks earnings forecasts

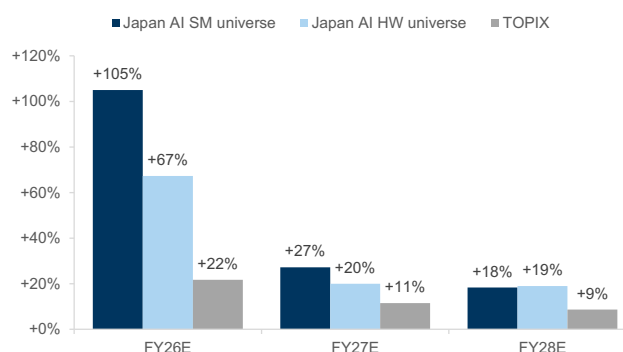


Earnings forecast revision index = (no. of analyst upward revisions for past 1 mo. - no. of downward revisions) Total no. of forecasts

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 8: AI HW and SM stocks are expected to deliver strong EPS growth

Consensus EPS growth

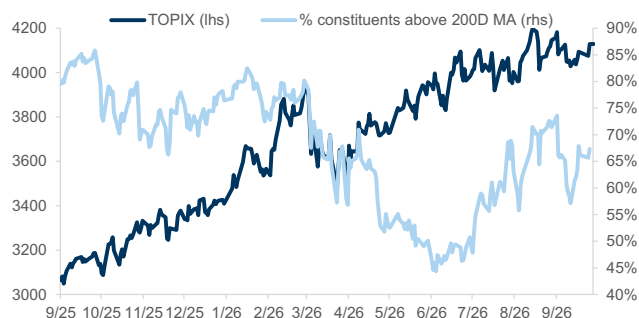


Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

In terms of market breadth, TOPIX performance did become very concentrated going into the end of June this year, with only 44% of TOPIX constituents trading above their 200D moving averages ([Exhibit 9](#)). However, market performance has broadened out since then. TOPIX market breadth is still quite narrow compared to historical averages, but not as concentrated as it was earlier this year ([Exhibit 10](#)).

Exhibit 9: 67% of companies trade above their 200D moving average vs 44% in June

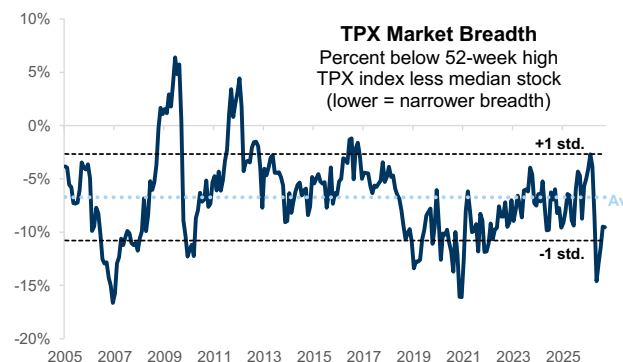
Data as of 25 September



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 10: Market breadth is still narrow compared to historical levels, but not as concentrated as it was earlier this year

Data as of 25 September



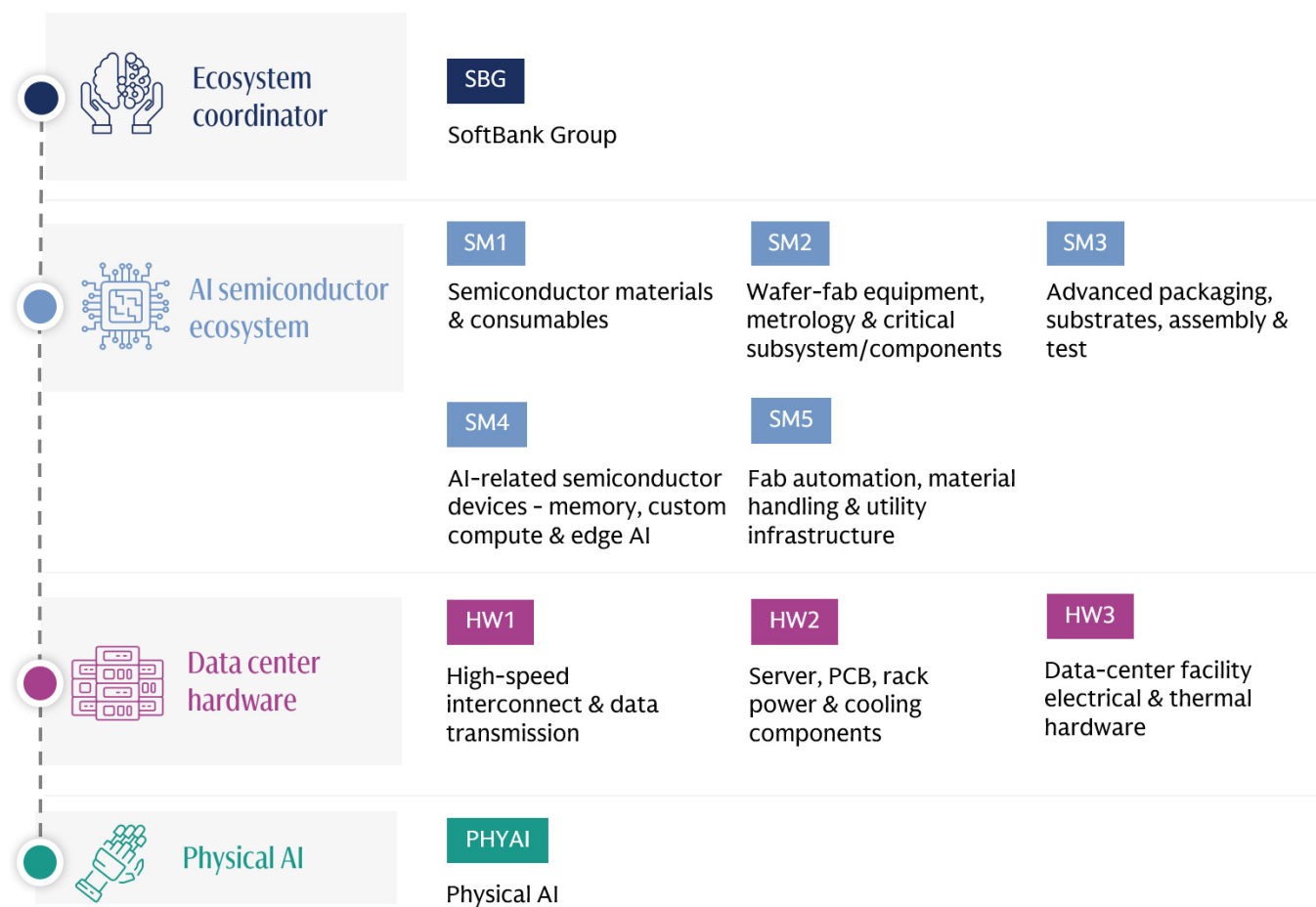
Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

In terms of market cap and liquidity, our universe includes AI-related stocks that have a market cap of at least ¥50bn and 6M advt of at least US\$5mn. We believe that this approach should help identify attractive AI-related SMID names with earnings profiles that could be highly geared to current and future AI demand. Fundamental exposure levels for these stocks are shown in the Appendix section of this report. There are currently 123 unique stocks in our AI SM/HW universe, with a combined 6M advt of US\$38bn.

Mapping and categorizing Japan's AI Value Chain

Our framework categorizes Japan's AI-exposed stocks based on each company's position within the broader AI value chain, linking individual products and businesses to distinct sources of AI-related demand. The two core groupings are the **AI Semiconductor Ecosystem (SM)** and **AI Data-Center Hardware (HW)**. SM can be broken down into areas such as materials, manufacturing equipment, advanced packaging and testing, compute and memory devices, and semiconductor-factory automation. HW includes areas such as high-speed connectivity, server- and rack-level components, and facility-level power and cooling. These categories provide a basis for comparing earnings trends, valuation differentials and price performance dispersion within the Japan AI universe. Differences between these various sub-layers should create opportunities for fundamental stock pickers. We assign more than one AI ecosystem classification to a company if its AI-related product offerings span across multiple layers. The Japanese AI ecosystem is shown in [Exhibit 11](#) below, with example companies by sub-layer shown in [Exhibit 12](#) on the next page. Alongside these core groupings, we identify **SoftBank Group** as a stand-alone *AI ecosystem coordinator*, and highlight Physical AI as a complementary investment thematic encompassing AI-enabled robotics and autonomous systems.

Exhibit 11: Japan AI ecosystem framework



Source: Goldman Sachs Global Investment Research

Exhibit 12: AI SM/HW layers and top market cap example companies

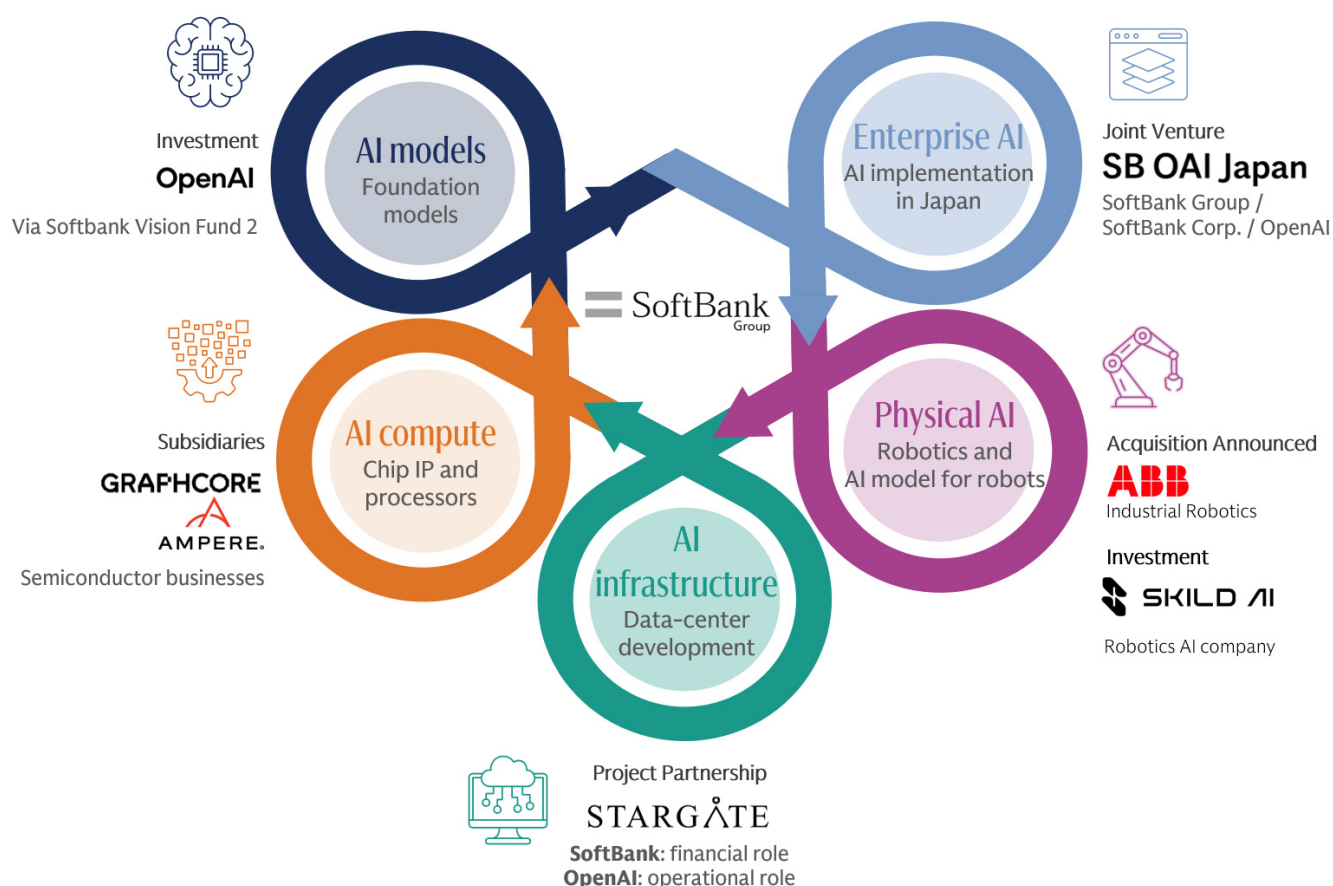
Source: Company data, Data compiled by Goldman Sachs Global Investment Research

SBG: SoftBank Group - the AI ecosystem coordinator

We believe that SoftBank Group (9984.T) represents what we refer to here as a stand-alone *AI ecosystem coordinator* which captures the coordination of capital, technology and deployment across the AI value chain. The company leverages this exposure via its investment in OpenAI (unlisted), its semiconductor businesses led by Arm Holdings (ARM) and its participation in infrastructure development at Stargate (unlisted). These activities connect AI models with computing capabilities and the infrastructure required for their deployment. In our opinion, the unique combination of investment ownership and strategic partnerships across multiple layers supports a distinct classification alongside our semiconductor and data-center hardware layers.

This ecosystem also extends into enterprise adoption and Physical AI. The SB OAI Japan partnership connects OpenAI's technology with implementation capabilities and Japanese enterprise customers, while robotics investments provide links to the development and deployment of intelligent machines. SoftBank is currently in the process of acquiring ABB's industrial robotics business, pending regulatory approvals. From a strategy perspective, this AI ecosystem coordinator layer appears to provide Japanese equities market investors with access to global AI platforms, their supply chains, and their continued commercialization.

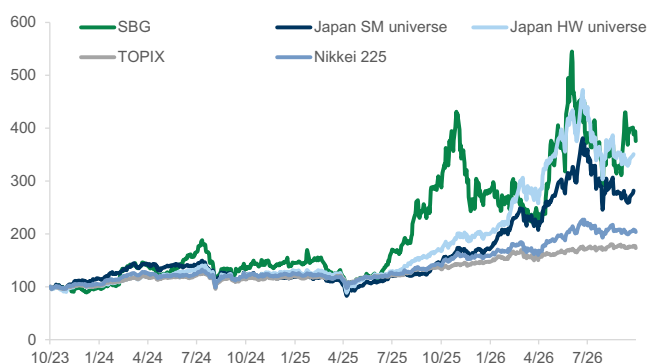
Exhibit 13: SoftBank Group's exposure to various layers of AI ecosystem and examples of investments and subsidiaries



Source: Company data

Recent share-price moves in SoftBank Group have been extremely volatile, and have tended to coincide with significant developments either in its major AI-related holdings, or around AI sentiment globally. For example, SoftBank Group shares rose +11% on 8 February 2024, following **Arm Holdings'** upbeat revenue outlook, but fell -8% on 27 January 2025 during the global AI equity selloff associated with **DeepSeek**. These episodes illustrate both SoftBank Group's upside leverage and downside vulnerability to company-specific news-flow and industry-wide AI re-pricing. As can be seen in [Exhibit 14](#), SoftBank tends to significantly outperform our broader SM/HW universes during periods of market exuberance on the AI thematic, but with more intense corrections.

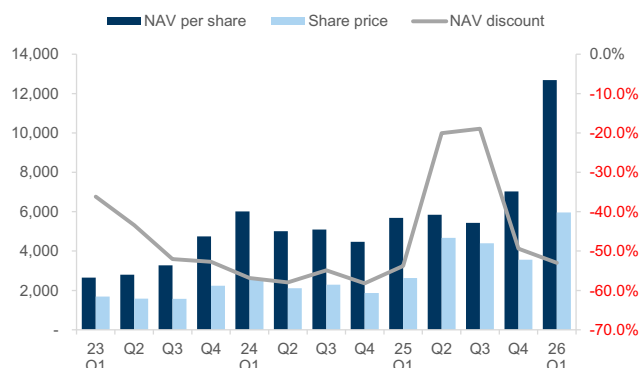
Exhibit 14: SoftBank Group outperformed broader AI SM/HW baskets in 2025 AI stock rally
Data as of Sep 25, 2026



SM/HW universe based on semi-annual rebalance using average of market cap and liquidity

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 15: SoftBank Group traded with large NAV discounts in recent years



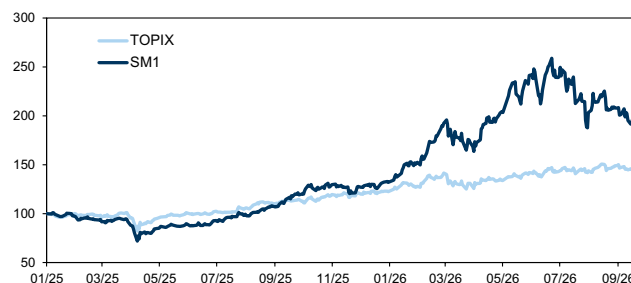
Source: Company filings, Data compiled by Goldman Sachs Global Investment Research

SM1: Semiconductor materials and consumables

Ticker	Name	Mcap (JPY bn)	Ticker	Name	Mcap (JPY bn)
4063	SHIN-ETSU CHEMICAL	11,690	4043	TOKUYAMA CORP	304
7741	HOYA CORP	7,864	5384	FUJIMI INC	278
5016	JX ADVANCED METALS	3,374	7826	FURUYA METAL CO LT	297
4004	RESONAC HOLDINGS C	2,860	4099	SHIKOKU KASEI HOLD	200
5201	AGC INC	1,242	5331	NORITAKE CO LTD	209
3436	SUMCO CORPORATION	1,091	3445	RS TECHNOLOGIES CO	163
4021	NISSAN CHEMICAL CO	1,124	4047	KANTO DENKA KOGYO	138
4186	TOKYO OHKA KOGYO	1,068	3104	FUJIBO HOLDINGS	113
4042	TOSOH CORP	864	4187	OSAKA ORGANIC CHEM	95
4182	MITSUBISHI GAS CHM	827	4369	TRI CHEMICAL LABOR	97
6890	FERROTEC CORP	575	4970	TOYO GOSEI CO LTD	81
4401	ADEKA CORPORATION	460	4022	RASA INDUSTRIES	75
5301	TOKAI CARBON CO	495			
429A	TEKSCEND PHOTOMASK	386			
4368	FUSO CHEMICAL CO	326			

Mcap as of 25 September, GS covered names highlighted

Source: FactSet, Goldman Sachs Global Investment Research



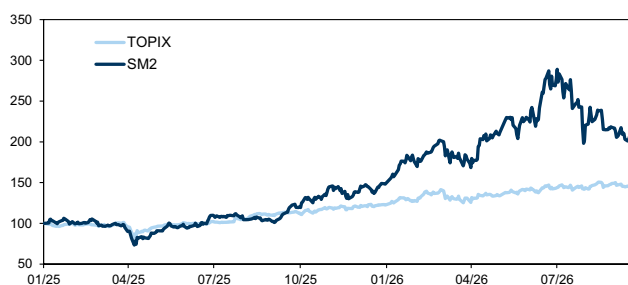
This layer covers semiconductor materials and consumables used primarily in the wafer fabrication of AI-related logic and memory chips. These include silicon wafers, photoresists, photomask blanks, high-purity process gases and chemicals, and chemical mechanical planarization (CMP) slurries and pads. This layer captures upstream inputs used to form, pattern, clean and planarize semiconductor structures. Materials for advanced packaging and assembly, including package substrates, build-up films and encapsulants, are classified in SM3 below according to their function.

Our GIR Japan Chemicals analysts are constructive on the silicon wafer sector, with AI server demand supporting strong volume growth in 300mm wafers, and the positive outlook for price increases in the upcoming round of LTAs (long-term agreements) ([LINK](#)). They think that global demand for 300mm wafers might expand beyond GPUs and HBM into CPUs, LPDDR (low-power double data rate) memory, enterprise SSDs, and power semiconductors. Based on this higher demand, they see the industry being at near full utilization in 2027, and expect considerable tightness in 2028 ([LINK](#)). In July, Micron signed a 10-year supply and \$500mn strategic financing agreement with GlobalWafers showing that customers are moving upstream to secure wafer capacity before a far tighter supply environment emerges ([LINK](#)).

SM2: Wafer-fab equipment, metrology & critical subsystems/components

Ticker	Name	Mcap (JPY bn)
8035	TOKYO ELECTRON	26,453
6273	SMC CORP	4,447
6920	LASERTEC CORP	3,857
7735	SCREEN HOLDINGS CO	2,621
6525	KOKUSAI ELECTRIC C	2,209
6361	EBARA CORP	2,031
5333	NGK CORP	1,549
6856	HORIBA LTD	1,007
5332	TOTO LTD	984
5344	MARUWA CO LTD	688
5393	NICHIAS CORP	650
6890	FERROTEC CORP	575
268A	RIGAKU HOLDINGS CO	378
6728	ULVAC INC	376
6407	CKD CORP	358

Ticker	Name	Mcap (JPY bn)
6951	JEOL LTD	396
6925	USHIO INC	335
6622	DAIHEN CORP	322
6590	SHIBAURA MECHATRON	264
6490	PILLAR CORPORATION	259
5310	TOYO TANSO CO LTD	159
6387	SAMCO INC	64
6266	TAZMO CO LTD	56

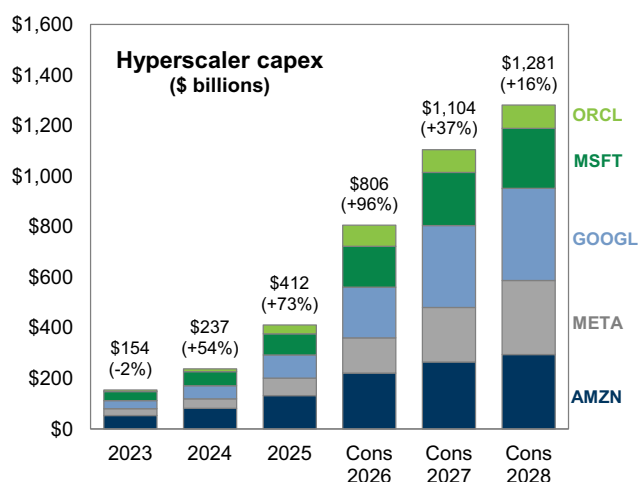


Mcap as of 25 September, GS covered names highlighted

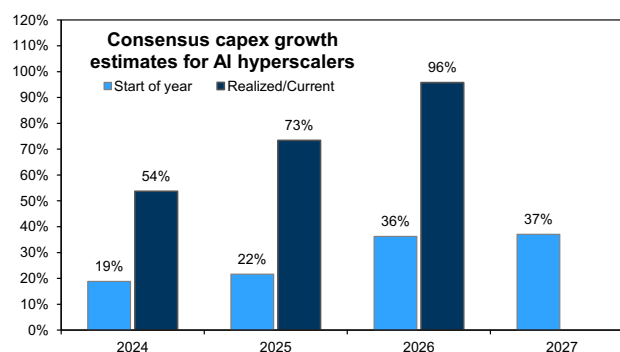
Source: FactSet, Goldman Sachs Global Investment Research

This layer covers equipment directly involved in semiconductor fabrication and process control. Examples include deposition, etching, cleaning, lithography, wafer processing, inspection and metrology equipment, together with critical subsystems such as vacuum equipment and radio-frequency power supplies. The layer also includes the tools and components supporting chip manufacturing and production yield. Equipment dedicated to packaging, assembly or electrical testing is classified in SM3, while fab-wide transport, wafer handling and general automation are assigned to SM5.

Share price performance in front-end semiconductor equipment stocks is closely related to hyperscaler capex expectations. Consensus expectations for 2026/27/28 capex spending are \$800bn/\$1.1bn/\$1.3bn respectively, with GS equity analysts expecting \$1.2tn of spending in 2027, and \$1.4tn of spending in 2028. Our US portfolio strategy team believes that capex spending will surpass consensus, but expect the rate of growth and upward surprises to slow down ([LINK](#), [LINK](#)). If AI capex does continue to trend higher, the most direct earnings beneficiaries should be infrastructure companies, of which Japanese front-end SPE companies (SM2) are major players.

Exhibit 16: Consensus hyperscaler capex estimates

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 17: Consensus hyperscaler capex estimates versus realized capex

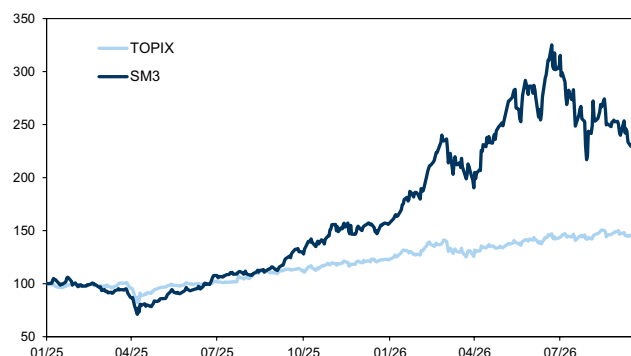
Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Our global tech analysts have raised 2026/27/28 Chinese semis capex forecasts from \$45/\$45/\$45bn to \$51/\$59/\$68bn. They model Chinese 2026-2028E WFE spending to account for 29-31% of global spending, but see the supply chain becoming more self-sufficient. With growing Chinese self-sufficiency, our Japan SPE analysts expect to see Chinese revenues grow in absolute levels but shrink as a ratio of sales due to other regional sales growing at a faster rate ([LINK](#), [LINK](#)).

SM3: Advanced packaging, substrates, assembly & testing

Ticker	Name	Mcap (JPY bn)
8035	TOKYO ELECTRON	26,453
6857	ADVANTEST CORP	24,888
4062	IBIDEN CO LTD	6,621
6146	DISCO CORPORATION	5,884
2802	AJINOMOTO CO INC	4,939
4004	RESONAC HOLDINGS C	2,860
3407	ASAHI KASEI CORP	2,247
6383	DAIFUKU CO LTD	2,108
5334	NITERRA CO LTD	1,518
7911	TOPPAN HOLDINGS IN	1,449
5706	MITSUI KINZOKU CO	1,319
4021	NISSAN CHEMICAL CO	1,124
4186	TOKYO OHKA KOGYO	1,068
4182	MITSUBISHI GAS CHM	827
7729	TOKYO SEIMITSU CO	747
4203	SUMITOMO BAKELITE	722
6787	MEIKO ELECTRONICS	640
3110	NITTO BOSEKI CO	562
4626	TAIYO HLDG CO LTD	558
6871	MICRONICS JAPAN CO	554

Ticker	Name	Mcap (JPY bn)
6754	ANRITSU CORP	462
5805	SWCC CORPORATION	417
7966	LINTEC CORP	417
4966	C. UYEMURA & CO	363
6925	USHIO INC	335
4061	DENKA COMPANY LTD	297
6590	SHIBaura MECHATRON	264
4099	SHIKOKU KASEI HOLD	200
6941	YAMAICHI ELECTRONI	176
6315	TOWA CORP	165
4971	MEC COMPANY LTD	132
6227	AIMECHATEC LTD	125
6855	JAPAN ELECTRON MAT	111
6627	TERA PROBE INC	109
6800	YOKOWO CO LTD	106
6961	ENPLAS CORP	96
6266	TAZMO CO LTD	56



Mcap as of 25 September, GS covered names highlighted

Source: FactSet, Goldman Sachs Global Investment Research

This layer covers the materials, equipment and interfaces used to assemble, connect and test semiconductor devices. Examples include IC package substrates, build-up films, encapsulants, glass substrates for packaging, bonding and dicing equipment, automated test equipment, wafer probers, probe cards and test sockets. The scope includes electrical testing at both wafer and packaged-device stages. Package-level interconnection belongs here, while server and switch printed circuit boards and their materials are assigned to HW2.

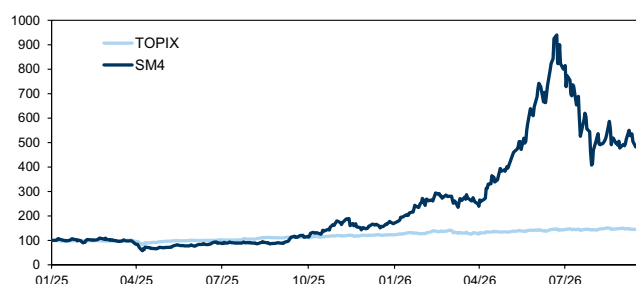
Share prices in many SM3 companies reflect supply/demand imbalances. Our global tech analysts ([LINK](#)) expect global server revenues to grow at +74%/+41%/+30% in FY26E-28E, and to reach US\$1.1tn. With high-end AI servers driving demand for high-end PCBs, our tech analysts expect Global PCB TAM to grow at +148% CAGR 2026-28E to US\$84bn, with shipments growing at +85% CAGR whilst ASP grows at +34% CAGR. Similarly, they expect the Copper Clad Laminate (CCL) market to grow at 161% 2026-2028E CAGR to reach US\$48bn by 2028, with CCL shipments growing at +77% CAGR while ASP grows at +48% CAGR ([LINK](#)). Our Chemicals analysts expect that the strong demand will be beneficial for the Japanese materials makers that supply raw materials for CCLs ([LINK](#)).

On ABF substrates, our industry analysts expect TAM to grow at +62% CAGR to US\$33bn in 2028, forecasting that the S/D gap for ABF substrates will widen from an 8% undersupply in 2026 to 51% in 2028 ([LINK](#)), which may allow an even further accelerated pace of price increases.

On SM3 equipment, our SPE analysts have noted that while expectations for equipment demand growth in CY26-27 may have been priced in, they expect earnings expectations to continue to rise for FY27-28 on WFE market growth ([LINK](#)). However, whilst the demand for products is strong, due to rising SPE component prices, our SPE analysts believe that preparedness for surges in demand, product competitiveness, and pricing strategies will be important ([LINK](#)).

SM4: AI-related semiconductor device including memory

Ticker	Name	Mcap (JPY bn)
285A	KIOXIA HLDGS CORP	30,603
6723	RENESAS ELECTRONIC	6,506
6526	SOCIONEXT INC	404



Mcaps as of 25 September, GS covered names highlighted

Source: FactSet, Goldman Sachs Global Investment Research

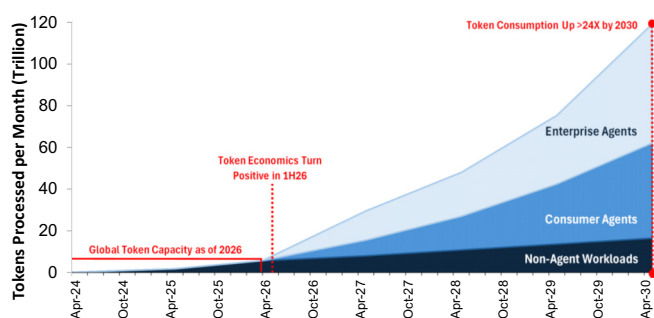
This layer includes devices that perform AI-related computation or provide semiconductor memory and storage. These include AI accelerators, custom ASICs and system-on-chips, dedicated edge-AI processors, high-bandwidth memory and NAND flash.

Recent weakness in memory stocks reflects concerns that the memory up-cycle will prove short-lived, amplified by leveraged positioning and momentum-driven selling. We share our [Asia strategy](#) and [Asia technology hardware teams'](#) view that this cycle is likely to be stronger and longer than previous upturns, supported by sustained AI demand and unusually tight supply.

Our global semiconductor analysts expect token consumption to increase 24-fold to 120 quadrillion tokens per month by 2030, driven by the shift toward consumer and enterprise AI agents. Hyperscaler capex is expected to reach US\$800bn in 2026, US\$1.1tn in 2027, and US\$1.3tn in 2028, with cumulative spending of approximately US\$6.0tn between 2026 and 2030. As can be seen in [Exhibit 18](#), the expansion of compute demand and infrastructure investment should support a sustained increase in memory and storage requirements. ([LINK](#))

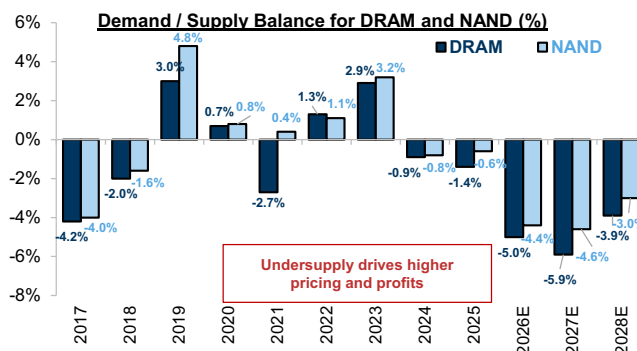
Our semiconductor analysts expect record DRAM and NAND undersupply in 2026, a deeper shortfall in 2027, and continued tightness through 2028. As can be seen in [Exhibit 19](#), this would represent a longer supply-constrained period than in previous cycles, supporting pricing and profitability through the operating leverage inherent in memory manufacturing. The relevant debate is therefore not only the near-term improvement in earnings, but also whether the market is underestimating its duration ([LINK](#)).

Exhibit 18: Global token consumption could increase 24x above current usage by 2030 driven by consumer and enterprise AI agents



Source: Data compiled by Goldman Sachs Global Investment Research

Exhibit 19: Record memory undersupply is likely to intensify further in 2027 and remain well short of demand in 2028 and likely in 2029-30 as well



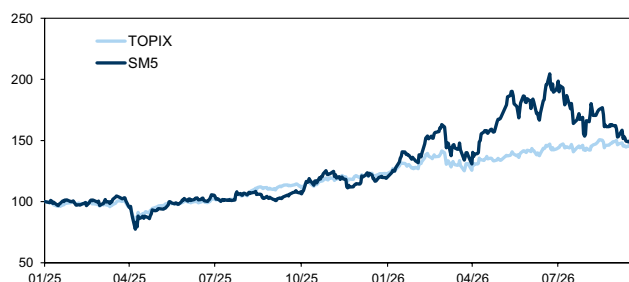
Source: Goldman Sachs Global Investment Research, Company Data

According to our GIR Japan semiconductor analyst, Kioxia (285A) management expects NAND demand from inference and agentic AI to remain strong and exceed supply through CY27, despite these applications still being at an early stage. Management is also progressing toward long-term agreements covering 50% of CY28 shipment volumes, with some discussions already extending beyond CY28.

SM5: Fab automation, material handling & utility infrastructure

Ticker	Name	Mcap (JPY bn)
6861	KEYENCE CORP	19,359
6273	SMC CORP	4,447
7012	KAWASAKI HEAVY IND	2,112
6383	DAIFUKU CO LTD	2,108
6645	OMRON CORP	1,304
6506	YASKAWA ELEC CORP	1,195
6370	KURITA WATER INDS	897
6481	THK CO LTD	766
6323	RORZE CORP	668
6368	ORGANO CORP	588
6507	SINFONIA TECHNOLOG	378
6622	DAIHEN CORP	322
6055	JAPAN MATERIAL CO	243
6254	NOMURA MICRO SCIEN	139

Ticker	Name	Mcap (JPY bn)
3449	TECHNOFLEX CORP	94
6266	TAZMO CO LTD	56



Mcap as of 25 September, GS covered names highlighted, performance is based on semi-annual rebalance using average of liquidity and market cap

Source: FactSet, Goldman Sachs Global Investment Research

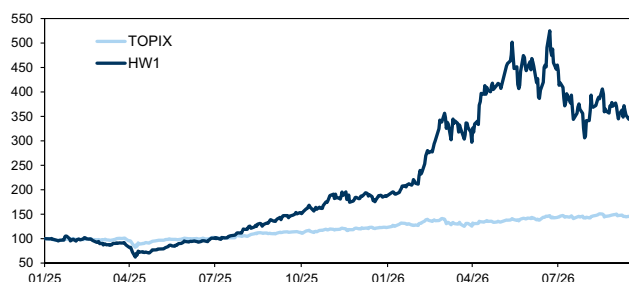
This layer covers systems that move materials and coordinate operations within semiconductor manufacturing facilities. Examples include clean room transport systems, wafer-handling equipment, vacuum transfer robots, and fab-related sensing, control and precision-positioning products. Inclusion requires identifiable semiconductor applications and meaningful exposure to semiconductor manufacturing investment.

On Factory Automation (FA) names in SM5, our GIR Japan machinery analysts believe that demand from AI / data center investment is strong in this cycle, and expect investors to focus on whether FA names can see price hike effects like those seen in other materials and components names. Of the FA products areas, our analysts expect electrical/electronic FA machinery and components that use memory/ICs to experience greater upward pressure than metal-fabricated products ([LINK](#), [LINK](#)).

HW1: High-speed interconnect & data transmission

Ticker	Name	Mcap (JPY bn)
6503	MITSUBISHI ELEC CP	11,071
5803	FUJIKURA	8,851
5802	SUMITOMO ELECTRIC	7,022
5016	JX ADVANCED METALS	3,374
5801	FURUKAWA ELECTRIC	2,704
5706	mitsui kinzoku co	1,319
6806	HIROSE ELECTRIC	864
6965	HAMAMATSU PHOTONIC	748
6890	FERROTEC CORP	575
4980	DEXERIALS CORP	538
6754	ANRITSU CORP	462
5805	SWCC CORPORATION	417
6834	SEIKOH GIKEN CO	284
6777	SANTEC HLDGS CORP	262

Ticker	Name	Mcap (JPY bn)
6941	YAMAICHI ELECTRONI	176
6779	NIHON DEMPA KOGYO	75



Mcap as of 25 September, GS covered names highlighted, performance is based on semi-annual rebalance using average of liquidity and market cap

Source: FactSet, Goldman Sachs Global Investment Research

This layer covers products that transmit data within and between AI computing systems and data centers. Examples include optical fiber and cables, optical connectors and transceivers, high-speed electrical interconnects, lasers, indium phosphide substrates, and communications test equipment. The layer groups these products by their role in data transmission, including optical semiconductor devices whose primary application is connectivity.

Price performance in HW1 stocks continue to be driven by AI data center-led supply demand tightness in a variety of products, including optical fiber cables, optical connectors, and optical devices. Earlier this year, our global tech analysts modeled total scale-up/scale-out demand from NVIDIA's GB300/Vera Rubin/Rubin Ultra rack

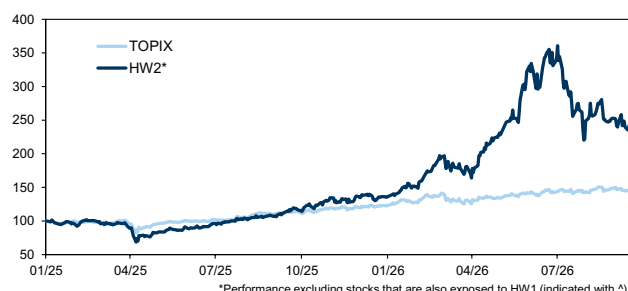
architecture. Scale out refers to the process of adding more servers horizontally, whilst scale up refers to the process of enhancing the performance of a single server. Our analysts model total demand for scale out/scale up to grow by 9x from US\$15.1bn in 2026 to US\$154.3bn in 2028 ([LINK](#)).

In another recent report, our global tech analysts raised their global optical transceiver outlook ([LINK](#)), and are now forecasting +33%/+81%/+115% 2026-28E TAM growth to US\$148bn, with 800G and above segments increasing at +69% CAGR to reach US\$130bn in 2028.

HW2: Server, PCB, rack power & cooling components

Ticker	Name	Mcap (JPY bn)
6981	MURATA MFG CO	15,747
6752	PANASONIC HLDGS CO	11,262
5803^	FUJIKURA	8,851
6723	RENEAS ELECTRONIC	6,506
6762	TDK CORP	5,789
6594	NIDEC CORPORATION	3,387
5016^	JX ADVANCED METALS	3,374
4004	RESONAC HOLDINGS C	2,860
5801^	FURUKAWA ELECTRIC	2,704
6963	ROHM CO LTD	1,935
6479	MINEBEA MITSUMI	1,535
5706^	mitsui kinzoku co	1,319
6976	TAIYO YUDEN CO LTD	1,260
6134	FUJI CORPORATION	729

Ticker	Name	Mcap (JPY bn)
5344	MARUWA CO LTD	688
6787	MEIKO ELECTRONICS	640
3110	NITTO BOSEKI CO	562
6278	UNION TOOL CO	283
6516	SANYO DENKI CO	211
5331	NORITAKE CO LTD	209
6996	NICHICON CORP	183
4461	DKS CO. LTD.	149
6707	SANKEN ELECTRIC CO	144
6999	KOA CORPORATION	86
3891	NIPPON KODOSHI	73
6997	NIPPON CHEMI-CON	73



Mcap as of 25 September, GS covered names highlighted, performance is based on semi-annual rebalance using average of liquidity and market cap

Source: FactSet, Goldman Sachs Global Investment Research

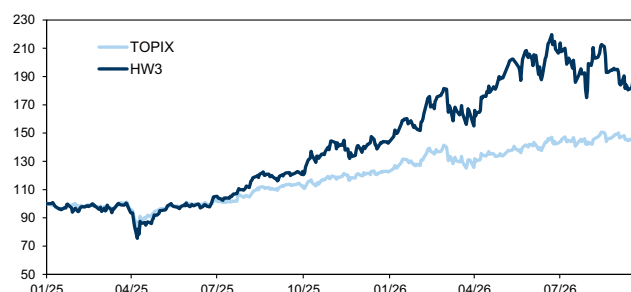
This layer covers equipment and components incorporated into, or directly supporting, servers and racks. Examples include servers, storage systems, server and switch printed circuit boards, electronic components, memory-interface devices, rack-level power supplies, cold plates, heat sinks and heat pipes. The layer captures system integration and the supporting electrical and thermal functions around compute and memory devices. Coolant distribution units and facility-level electrical and cooling systems are assigned to HW3 according to their infrastructure function, including when installed near racks.

Our global tech analysts expect total server revenues to grow at +47% CAGR in 2026-28E to US\$1.1tn, led by AI server racks, which they expect to grow at +118% CAGR ([LINK](#)). On cooling, our tech analysts expect Global Server Cooling TAM to grow to US\$18bn in 2027, with liquid cooling penetration to reach 80% ([LINK](#)).

Our global tech team expects global Battery Backup Unit (BBU) market size to grow at +87% 2025-2030E CAGR to US\$50bn in 2030, on higher global GPU power consumption and average BBU attach rates ([LINK](#)). At a higher end, they see a bull case where BBU TAM could grow at +103% CAGR to US\$75bn.

HW3: Data center power & thermal hardware

Ticker	Name	Mcap (JPY bn)
6501	HITACHI	25,109
6503	mitsubishi elec cp	11,071
6752	PANASONIC HLDGS CO	11,262
6594	NIDEC CORPORATION	3,387
6361	EBARA CORP	2,031
6504	FUJI ELECTRIC CO.	2,033
1969	TAKASAGO THML ENG	663
6674	GS YUASA CORP	542
6508	MEIDENSHA CORP	522
6622	DAIHEN CORP	322
6516	SANYO DENKI CO	211
6707	SANKEN ELECTRIC CO	144
6617	TAKAOKA TOKO CO LT	113



Mcap as of 25 September, GS covered names highlighted

Source: FactSet, Goldman Sachs Global Investment Research

This layer covers the electrical and thermal systems supporting data center operations. Examples include transformers, switchgear, power-distribution equipment, uninterruptible power supplies, backup generators, coolant distribution units, chillers, fan walls and air-conditioning systems. Inclusion emphasizes identifiable data center products and deliveries. General transmission cables and grid infrastructure sit outside this layer unless a distinct data-center application is established.

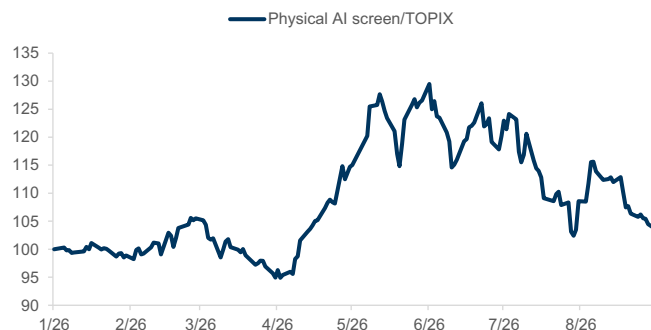
Our industry analysts have recently revised their global AI/data center power demand forecasts, and now assume +170% (previously +117%) global data center power demand growth in 2030 vs 2015 levels, and see over 60% of this demand coming from the US (see [Power demand, cyclical progression and Sustainability implications](#)).

Physical AI as a longer-term AI thematic, but faster adoption than previously assumed

As highlighted in [Five investment themes for equity investors in 2026](#) (5 January 2026), we had expected Physical AI to be one of the five major investment themes in Japan this year. However, the performance of our equal-weighted screen only noticeably outperformed briefly during April to May this year, and has been underperforming TOPIX since then. While Physical AI remains a longer-term structural theme given its modest near-term revenue contribution for select constituent companies, implementation progress is advancing faster than expected, supported by a stronger adoption outlook and increasing clarity around key strategic AI alliances.

Exhibit 20: Our previous equal-weighted Physical AI screen outperformed TOPIX during April to May, but subsequently underperformed

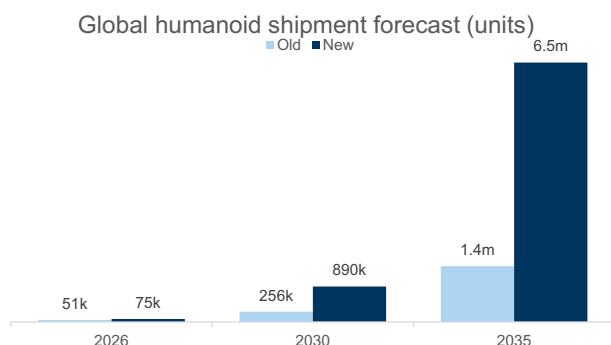
Equal-weighted, indexed Jan 2026 = 100



Source: FactSet, Goldman Sachs Global Investment Research

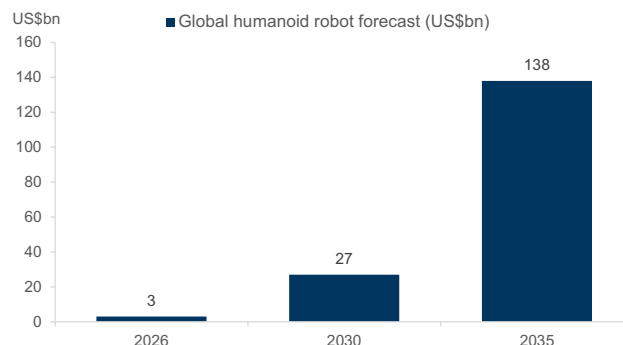
Our global research teams have also recently raised their humanoid market forecasts, and are now expecting the market to reach about 890k units in 2030 and 6.5mn units in 2035, up from prior 256k and 1.4mn. They expect the market to reach US\$138bn in 2035.

Exhibit 21: Our global research teams raised their humanoid market forecast, expecting to reach about 890k units in 2030 and 6.5m in 2035



Source: Goldman Sachs Global Investment Research

Exhibit 22: On a dollar basis, the market is expected to reach US\$138bn in 2035



Source: Goldman Sachs Global Investment Research

Other than humanoid robots, NVIDIA also provided a number of specific examples of Physical AI ([Exhibit 23](#)), such as AMRs, smart robot arms, autonomous vehicles, and smart spaces ([LINK](#)).

Exhibit 23: Examples of Physical AI as suggested by NVIDIA**Autonomous Mobile Robots (AMRs)**

AMRs are intelligent robots that navigate environments independently, using AI and onboard sensors (LiDAR, cameras) to perceive surroundings, avoid obstacles (including humans), and dynamically plan routes in real-time. This enables greater safety, precision, and adaptability in complex settings like warehouses for material transport and inspection.

**Manipulators/robot arms**

Robot arms are mechanical devices mimicking human arms, performing precise tasks like welding and assembly. Physical AI allows them to adjust grasping strength and position based on object pose, demonstrating fine and gross motor skills tailored to the object type, enhancing flexibility in manufacturing.

**Surgical robots**

Surgical robots assist surgeons with enhanced precision, control, and visualization for minimally invasive procedures. Physical AI enables them to learn intricate tasks such as threading needles and performing stitches, highlighting their adaptability and precision for specialized medical applications, leading to faster patient recovery.

**Humanoid robots / general-purpose robots**

Humanoid and general-purpose robots are designed to interact with human-centric environments. They require both gross and fine motor skills, along with the ability to perceive, understand, reason, and interact with the physical world to perform diverse tasks, from personal assistance to complex service roles.

**Autonomous Vehicles (AVs)**

Physical AI enables AVs to process real-time sensor data to perceive and understand their surroundings. Reasoning vision-language-action (VLA) models make informed decisions, while training in physically accurate simulations helps them detect pedestrians, respond to traffic/weather, and adapt to unexpected scenarios.

**Smart Spaces**

Physical AI enhances functionality and safety in large indoor/outdoor spaces like factories. Using fixed cameras and advanced computer vision, teams can optimize dynamic route planning, track entities, and improve operational efficiency. Video analytics AI agents detect anomalies and provide real-time alerts.



Source: Goldman Sachs Global Investment Research

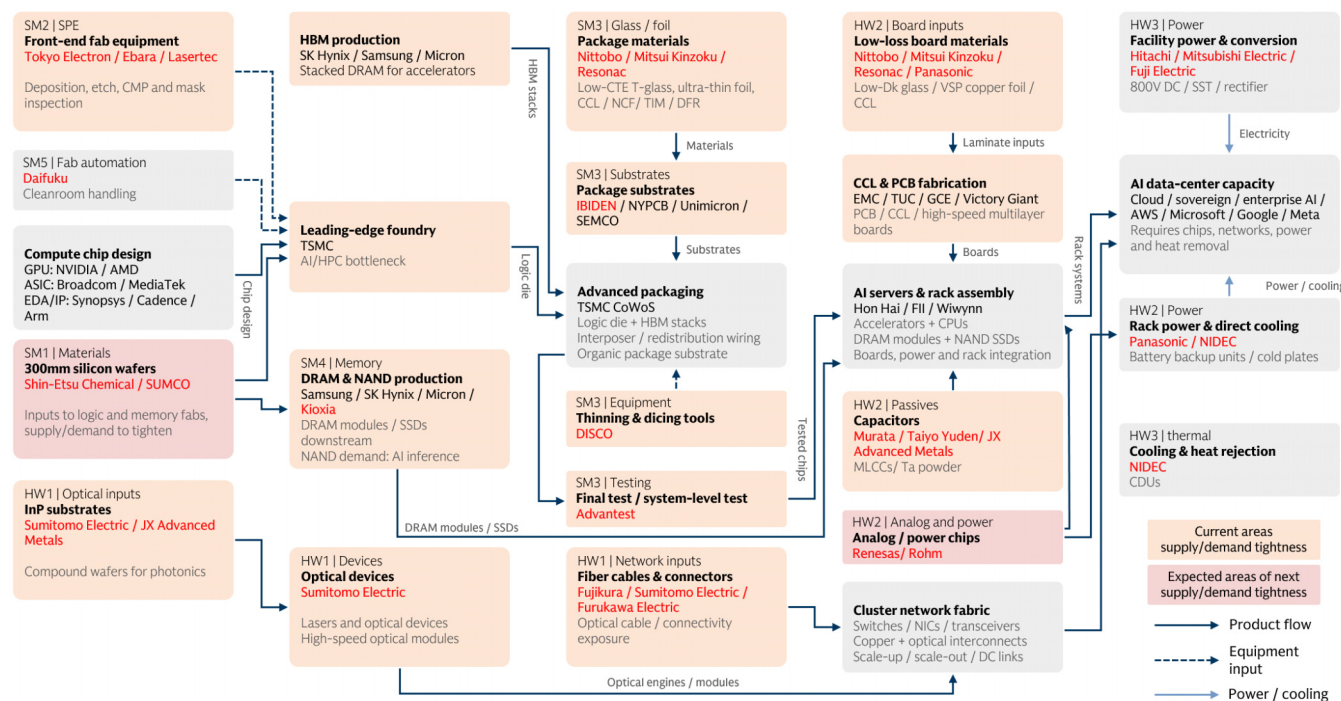
The Physical AI thematic is still in its early stages, but so far in CY2026 we have already started to see more positive developments. On 16 July, NVIDIA CEO Jensen Huang announced a series of collaborations with Japanese companies centered on physical AI (see [IT sector report](#), and [Machinery sector report](#)).

Japan's position in the global AI ecosystem

Japan's participation in global AI is anchored in specialized technologies across the supply chain. Japanese suppliers hold leading positions in selected semiconductor materials and manufacturing processes, including silicon wafers and coater/developer equipment. Their capabilities also extend across packaging, testing, optical components and thermal-management materials.

Japan's supplier base also overlaps with several areas of supply constraint in the AI value chain. Our Asia Technology team's report ([LINK](#)) highlighted tight supply-demand conditions extending beyond memory into niche electronic components and materials. Examples include low-thermal-expansion glass and ultra-thin copper foil for semiconductor packaging substrates, and low-dielectric-constant glass and very-low-profile copper foil for AI server motherboards. These areas connect Japanese suppliers to constraints spanning SM3 and HW2, alongside NAND-memory exposure in SM4. The report also identifies potential further tightening in silicon wafers and analog/power semiconductors. This overlap illustrates Japan's role in supplying specialized inputs required to expand AI infrastructure.

Exhibit 24: Japanese companies are involved in multiple bottlenecks in the AI value chain, and our sector analysts expect supply/demand tightness to expand

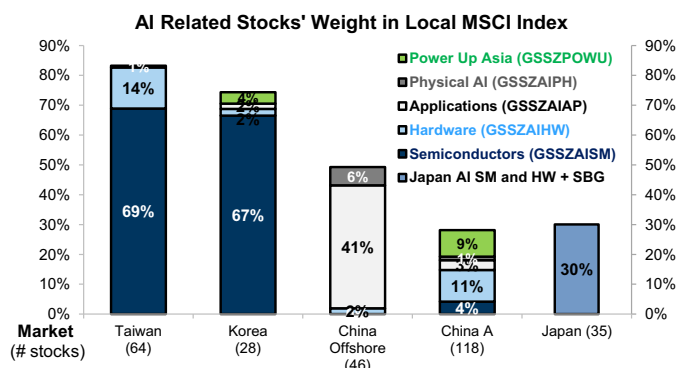


Japanese companies highlighted in red

Source: Goldman Sachs Global Investment Research

The composition of AI exposure is an important distinction across equity markets. Using comparable definitions, AI SM and HW-related companies account for 27% of MSCI Japan market capitalization (30% if also including SoftBank Group), versus 83% in Taiwan and 69% in Korea. China offshore is exposed largely to AI applications (41% of mcap), and China A is diversified across different AI themes (49% of mcap combined). It is important to note that AI-adjacent themes such as Power and Application are not included in this chart for Japan so the true exposure could be higher.

Exhibit 25: 30% of MSCI Japan by market cap are stocks in our AI (SM+HW+SBG) universe, Taiwan and Korea are more concentrated into SM alone, and China Offshore is exposed to Applications



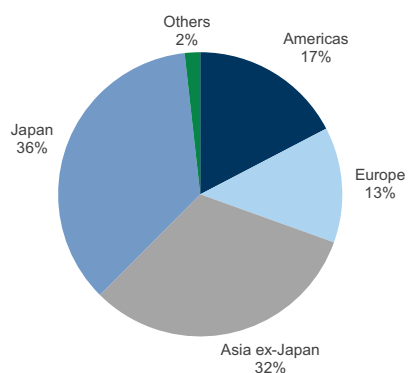
Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Geographic sales exposure connects the Japanese universe to overseas investment and production. As can be seen in [Exhibit 26](#), overseas markets account for 64% of aggregate reported sales among the universe's 124 constituents (SM+HW+SBG), with Asia ex-Japan representing the largest overseas contribution (32%). This suggests that AI-related stocks have much higher overseas sales ratio vs TOPIX, which has an overseas sales ratio of 43%, and Asia ex-Japan's sales ratio of 13%.

The sub-layer breakdown in [Exhibit 27](#) shows SM2 (wafer-fab equipment) has the highest sales exposure to Asia ex-Japan, SM4 (AI compute and memory) has the highest exposure to Europe, SM5 (fab-automation) has the highest exposure to Americas, and HW1 (data transmission) has the highest domestic sales exposure.

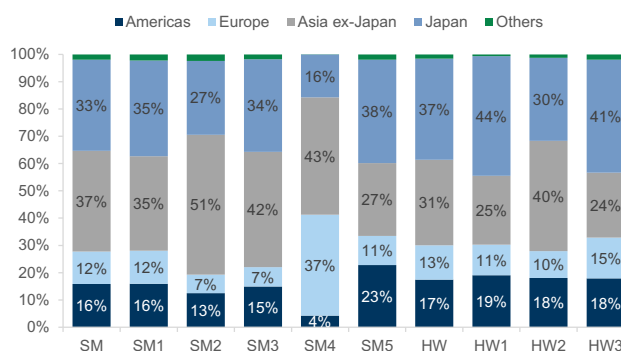
Exhibit 26: AI SM and HW companies have large sales exposure to overseas, especially Asia ex-Japan

FY0 sales exposure of AI semiconductor ecosystem and data center hardware stocks



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 27: Within the universe, SM2 has the highest sales exposure to Asia ex-Japan, SM4 has the highest exposure to Europe, SM5 has the highest exposure to Americas, and HW1 has the highest domestic sales exposure



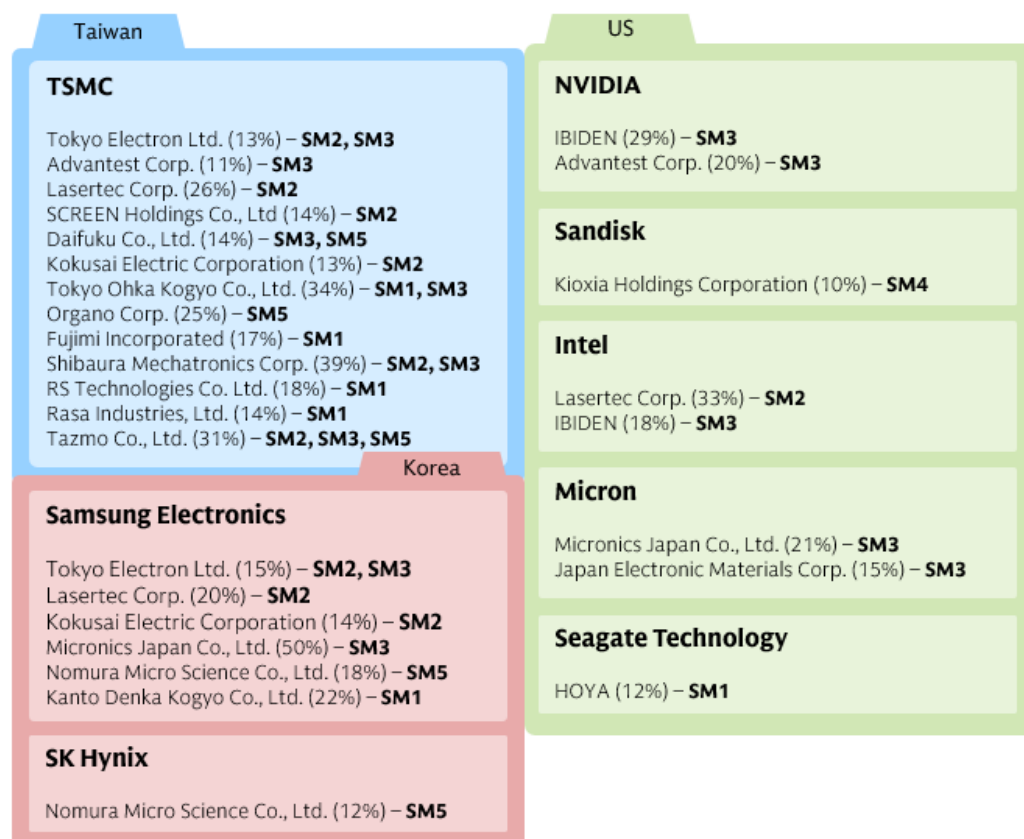
Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Disclosed customer relationships provide a more direct view of notable commercial connections. Our map of major customers reported in annual securities filings identifies links between Japanese suppliers and overseas buyers. These relationships help trace how Japanese companies participate in semiconductor manufacturing and the supply of computing infrastructure.

Exhibit 28 also highlights key customer exposure overseas for SM-related stocks. Annual disclosures suggest Japan AI SM companies are key suppliers to major players overseas including TSMC in Taiwan, Samsung Electronics and SK Hynix in Korea, and NVIDIA in the US.

Exhibit 28: Japanese companies provide inputs that are important for major overseas AI players

Major supplier-customer relationships involving Japan AI SM companies (>10% of sales)



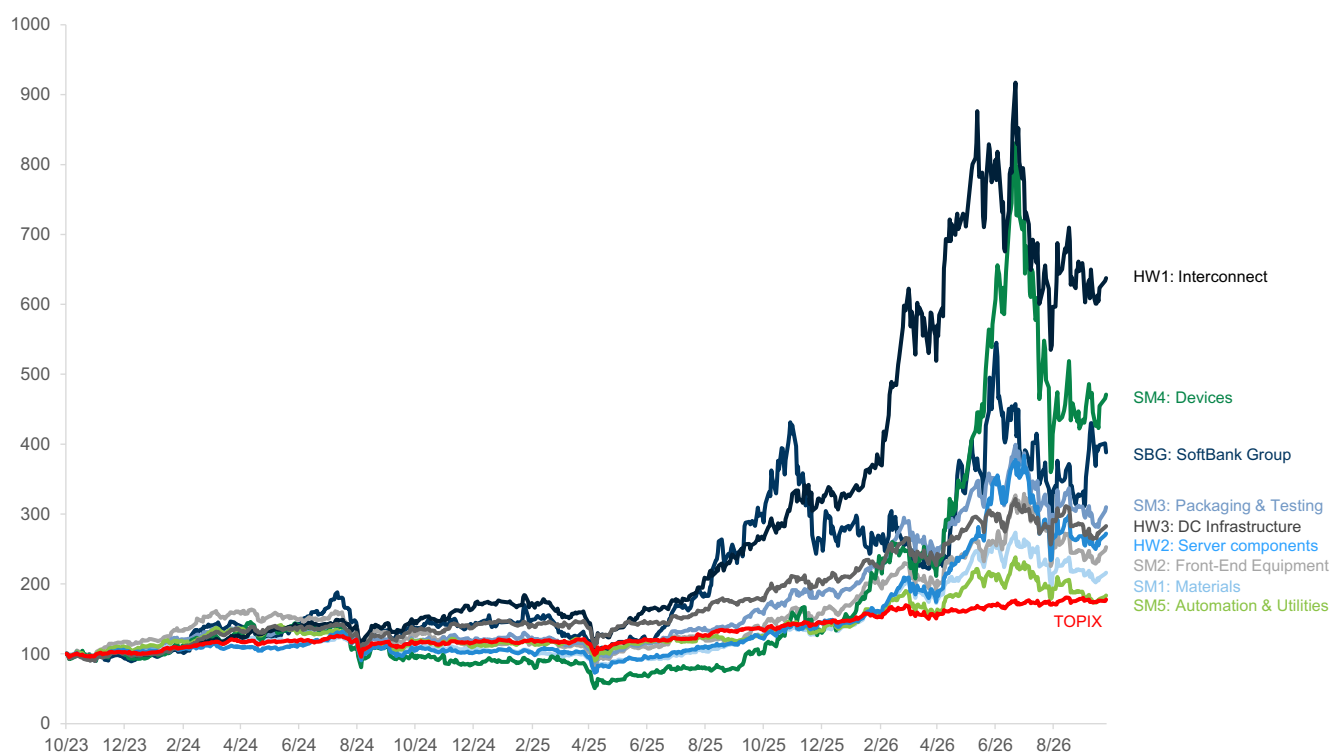
Percentage indicates contribution to sales

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

A more attractive risk/reward for Japan AI SM/HW

In this section, we revisit the price moves for various sub-layers in the AI value chain over the past three years, and examine which ones appear relatively more attractive now from a valuation standpoint. All the sub-layers in this section are re-balanced on a semi-annual basis based on liquidity and market capitalization. HW2 stocks that also have product exposure in HW1 (Fujikura, JX Advanced Metals, Furukawa Electric, Mitsui Kinzoku) are excluded in HW2 in this section given their large weights. We believe this provides a clearer performance and valuation analysis for the remaining stocks in HW2.

Exhibit 29: Performance of SBG and AI SM/HW layers



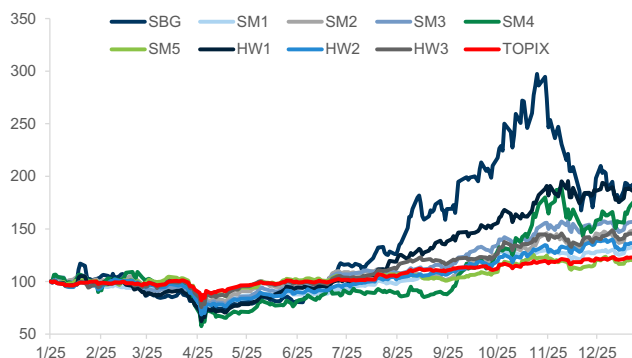
Rebalanced semi-annually based on average of liquidity and market cap

Source: FactSet, Goldman Sachs Global Investment Research

Revisiting the performance of various SM/HW layers in 2025, all layers except **fab-automation, material handling & utility infrastructure (SM5)** outperformed TOPIX as can be seen in [Exhibit 30](#). **SoftBank Group** rallied as much as +197% and peaked in late October, ending the year +92%. The best performing layer was **high-speed interconnect and data transmission (HW1)**, which ended the year +86%. **AI-related semiconductor device including memory (SM4)** also started to outperform from September and ended the year +68%. **Fab-automation, material handling & utility infrastructure (SM5)** was the worst performing layer, only up +19% in the year and underperformed TOPIX by 3ppts.

YTD in 2026, **AI-related semiconductor device including memory (SM4)** has been the best performing layer, up as much as +458% and peaked on June 22. This layer subsequently corrected by as much as -57%, bottoming on July 29. It is also note-worthy that the AI rally also broadened to **server, PCB, rack power & cooling components (HW2)** stocks, up as much as +168%, and remains one of the best performing layers YTD (+89%).

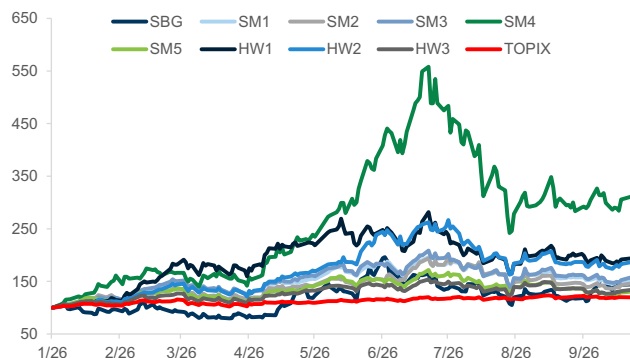
Exhibit 30: SoftBank Group and high-speed interconnect & data transmission (HW1) were the best performers in 2025



Rebalanced semi-annually based on average of liquidity and market cap

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 31: AI devices including memory (SM4) has been the best performer YTD, the rally also broadened out to HW2 stocks including MLCC related names
Data as of Sep 25, 2026

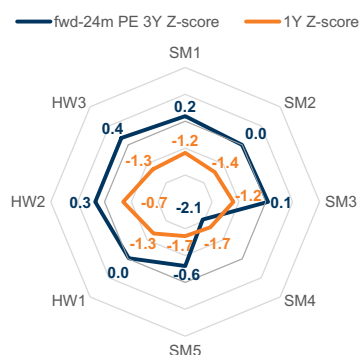


Rebalanced semi-annually based on average of liquidity and market cap

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

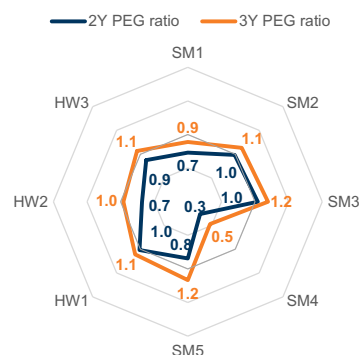
We analyze the recent valuations of each sub-layer within the AI universe. [Exhibit 32](#) suggests that after the correction since late June, valuations are now more attractive across all layers, with a fwd-24m (12 months period ending forward 24 months) PE 3Y z-score of close to 0, and deep in negative territory for past 1Y z-scores. Relative to earnings growth for next 2-3 years, [Exhibit 33](#) shows that the PEG ratio is also attractive. Particularly, SM1 and SM4 stocks have PEG ratios of below 1x. While low PEG ratios in these layers partly reflect uncertainty over durability of the current cycle, our analysts expect supply-demand tightness to persist in memory and potentially broaden to wafers, which may mitigate the typical earnings fade. We believe the valuation reset across layers has improved the risk-reward.

Exhibit 32: 3Y valuation z-scores are now close to 0 for most layers, and deep in negative territory for 1Y z-scores
As of 14 September 2026



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 33: AI SM/HW layers PEG ratios
Data as of 14 September 2026

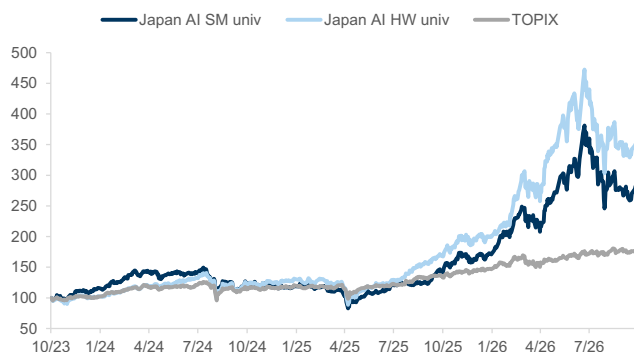


Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

The liquidity, valuations, EPS growth, and price returns for the broader **AI semiconductor ecosystem universe** and **AI data-center hardware universe** are shown in [Exhibit 35](#). Over the past 3 years, the semi-annually re-balanced SM universe returned +182% while the HW universe returned +251%, compared with +78% for TOPIX.

Exhibit 34: Japan AI SM universe and AI HW universe performance

Data as of Sep 25, 2026



Rebalanced semi-annually based on average of liquidity and market cap

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 35: Summary of our GS Japan AI SM and HW universe

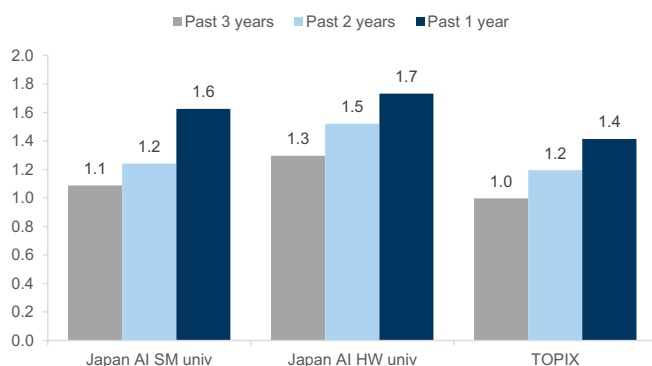
Data as of Sep 25, 2026

		JP SM	JP HW
Total liquidity	# of stocks	92	45
Total Listed Market Cap (¥bn)		208,751	124,366
Total 6M ADTV (\$mn)		27,201	10,814
Valuations (Aggregate)	FY27E PE	13.3	17.7
	FY26E PB	3.5	2.8
	Fwd 2Y PEG	0.6	0.4
	Fwd 3Y PEG	0.8	0.3
	fwd-24m PE z-score (1Y)	-2.0	-0.7
	fwd-24m PE z-score (3Y)	-1.6	0.5
EPS Growth (%)	FY26E	105%	67%
	FY27E	27%	20%
	FY28E	18%	19%
Price return	2024	2%	27%
	2025	44%	53%
	2026 (YTD)	65%	76%

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 36: Risk-adjusted returns have improved relative to TOPIX

Sharpe ratio

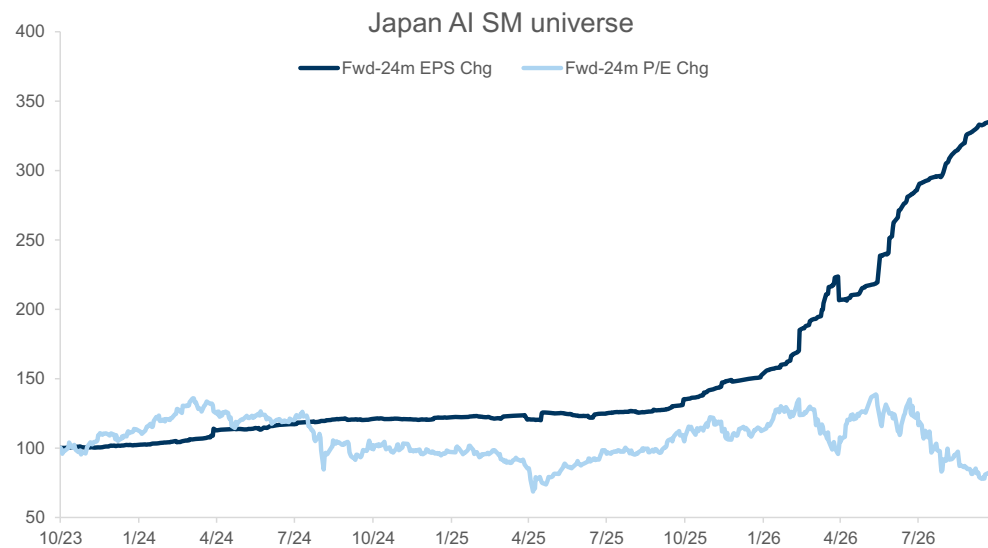


Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

In the past 3 years, semi-annually re-balanced Japan AI SM universe's +182% cumulative price return has been accompanied by a +236% change in fwd-24m forward EPS and -16% decline in forward P/E. Conversely, Japan AI HW universe's +251% cumulative price return has been accompanied by both +151% increase in fwd-24m EPS and +40% rise in forward PE. For performance breakdown of each layer, please see [Exhibit 74](#) in the Appendix section of this report.

Exhibit 37: Performance of AI SM universe has been driven by EPS revisions

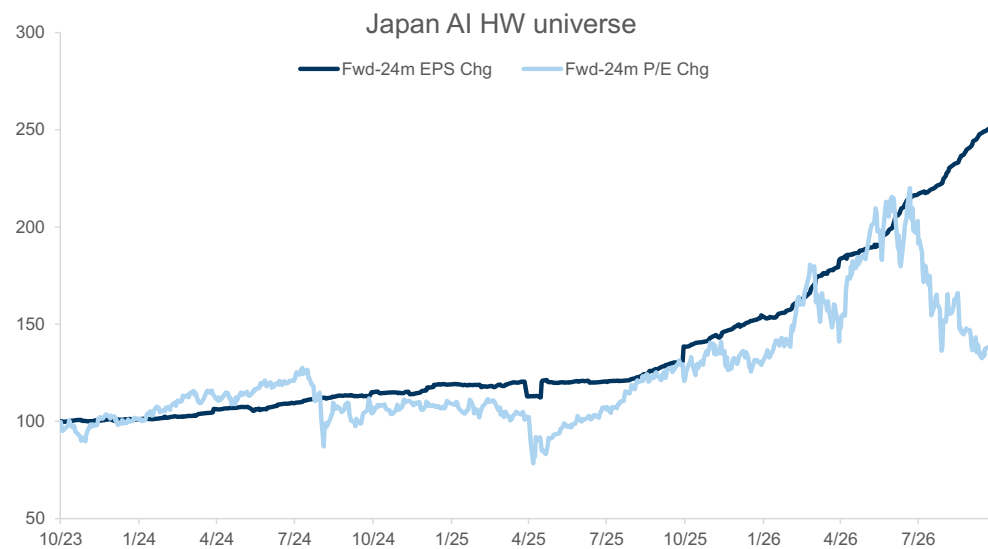
Data as of Sep 25, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 38: Performance of AI HW universe has been driven by both EPS and PE change

Data as of Sep 25, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Macro headwinds and tailwinds

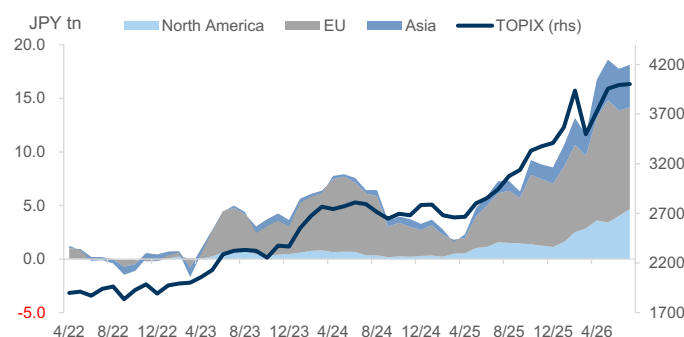
In this section, we address some of the broader macro concerns that could impact sentiment for AI stocks in Japan moving forward. Concerns about the negative impact on the AI thematic from broader macro factors such as a stronger yen, higher interest rates and geopolitics have been voiced frequently by clients in recent months

Stronger yen

As highlighted in the previous section about higher overseas exposure for AI-related stocks in Japan relative to TOPIX, we estimate AI-related companies' FX sensitivity using bottom-up company reported FX sensitivity data. AI-related stocks aggregate EPS is estimated to increase/decrease by **4.0%** for every 10Y rise/decline in USDJPY, vs **3.5%** for TOPIX. We think that the relatively small difference is due to the lack of FX-sensitive Autos sector names in the AI SM/HW universe.

While a stronger yen could negatively impact earnings growth of AI-related stocks, a gradual strengthening of the yen can improve the dollar-adjusted price returns for foreign investors. Particularly, foreign flows from North America have been stronger this year, coinciding with the strong performance of AI-related stocks ([Exhibit 39](#)). This suggests US-based investors were likely positioned into AI-related names in Japan. A sustained improvement in the dollar-adjusted price return for these stocks could attract further inflows from foreign investors.

Exhibit 39: Foreign flows from North America have totalled JPY3.5tn, contributing to 34% of total foreign inflows YTD
Data as of July 2026

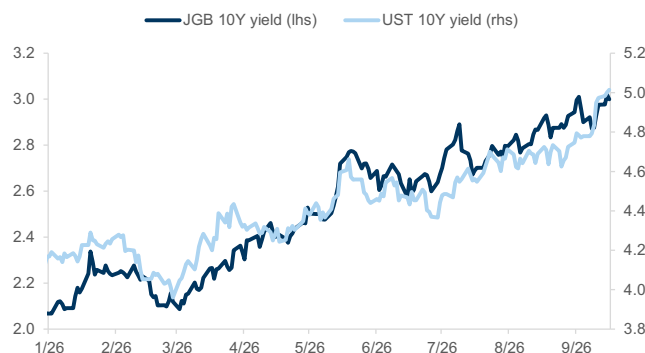


Source: JPX, Data compiled by Goldman Sachs Global Investment Research

Higher interest rates globally

With rising interest rates around the world including JGB 10Y yield reaching 3% and UST 10Y yield reaching 5%, the impact on equity valuation is a major concern for investors. As can be seen in [Exhibit 41](#), both our AI SM universe and AI HW universe experienced larger valuation declines during weeks of sudden US interest rate increases.

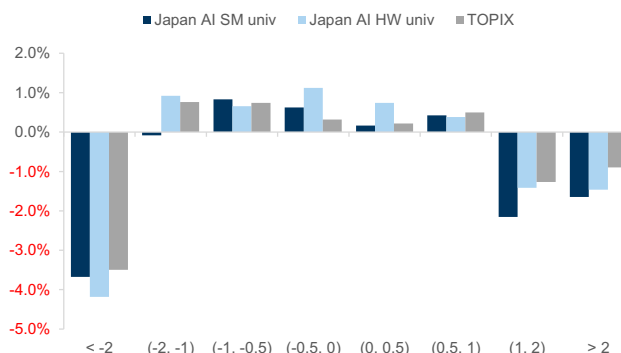
Exhibit 40: Rising global yields are a valuation headwind for equities



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 41: AI-related stock valuations are more sensitive to large increases in US yields

Average weekly STM PE change under different UST 10Y yield weekly change z-score ranges, past 3 years



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

However, aligned with our latest view on TOPIX ([LINK](#)), we believe that earnings growth is the most important driver of Japanese equities including AI-related stocks. While we see more room for valuation recovery for SM segment of our AI universe, we think that if the next earnings season in late October provides further evidence of strong earnings fundamentals, the recovery in AI-stock share prices should also continue. Furthermore, if our expectation is correct, it could also help overcome the valuation headwind, especially when the pace of rate increases normalizes.

Policy support - Takaichi administration's 17 strategic sectors

The **Takaichi administration's 17 strategic sectors** ([LINK](#)) span several parts of our framework, including AI and semiconductors, digital and cybersecurity, and information and communications. Priorities include **semiconductor production, Physical AI, domestic data centers and power systems, and optical networks**. Together, these measures provide potential support through investment, technology development and domestic adoption across Japan's AI semiconductor and hardware ecosystems. For detailed summary of key products and targets, see our Japan Economics team's report ([LINK](#)).

Exhibit 42: Policy supports multiple areas of AI development directly and indirectly

Policy-to-theme mapping across five selected strategic sectors

Selected policy priority	SM Ecosystem	HW1 Connectivity	HW2 Server / rack	HW3 Facilities	Physical AI	Cybersecurity
AI and Semiconductors	Direct	-	Enabling	-	Direct	-
Digital and cyber security: cloud & DCs	Enabling	Direct	Direct	Direct	-	-
Information and Communications	-	Direct	-	-	-	-
Materials (critical minerals, parts and materials)	Direct	Enabling	Enabling	-	Enabling	-
Resources, energy security and green transformation (GX)	Enabling	-	-	Enabling	-	-
Digital / cyber: cybersecurity	-	-	-	-	-	Direct

Source: Cabinet Office, Goldman Sachs Global Investment Research

Cybersecurity represents a complementary dimension of AI-related investment. Japan's growth strategy explicitly prioritizes security products and services suited to the AI era, including support for domestic technology development and adoption. The government targets annual revenue exceeding ¥3tn for domestic cybersecurity companies by 2035. These initiatives provide a policy basis for examining Japanese security providers alongside the companies supplying AI infrastructure.

Recent debate over the pace of frontier AI development has further highlighted security requirements. Anthropic's September 2026 threat report documented malicious actors using AI to automate parts of their operations, while CEO Dario Amodei called for stronger safeguards and independent evaluation alongside more measured capability advancement. We therefore include a separate **Japan cybersecurity screen** covering companies involved in supporting secure AI adoption ([Exhibit 63](#)).

Physical AI - labor scarcity strengthens the incentive to automate

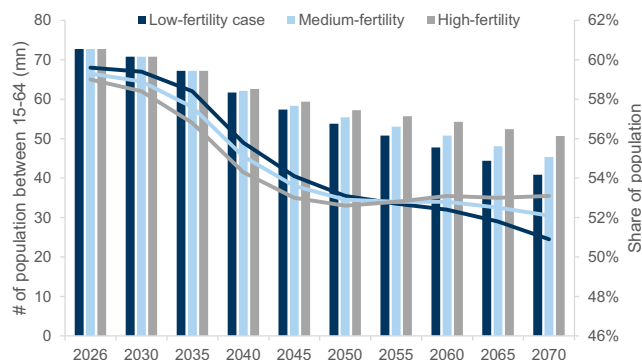
Japan's demographic outlook provides a structural source of demand for Physical AI. Under the National Institute of Population and Social Security Research's 2023 medium projection, the population aged 15–64 is expected to fall from 75.1 million in 2020 to 62.1 million by 2040, a decline of approximately 17% ([LINK](#)). This demographic pressure reinforces the need to increase output per worker. We see this as a long-term incentive for investment in technologies that help companies sustain operations with fewer available workers.

As can be seen in [Exhibit 44](#), reported labor expenses for TOPIX companies excluding Financials reached 10.5% of sales in FY2025, up from 8.5% in FY2016. The latest reading is the highest level in the past 10 years excluding denominator effect during COVID–19.

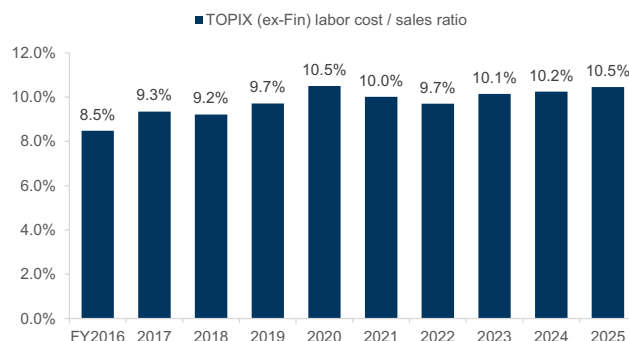
Our sector analysts forecast humanoid robot unit prices to decline from approximately US\$42,000 in 2025 to US\$21,000 by 2035 and could bring about a crossover point with rising labor costs ([LINK](#)).

Exhibit 43: Demographic pressure reinforces the need to increase output per worker

Number of population aged 15-64 (column, lhs), and % of population (line, rhs)



Source: IPSS

Exhibit 44: Labor costs absorbing a growing share of corporate revenues


Source: Toyo Keizai, FactSet, Data compiled by Goldman Sachs Global Investment Research

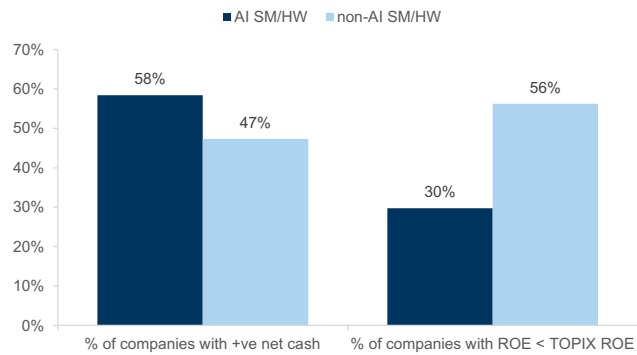
Government support could also help accelerate humanoid robot adoption. Japan's AI robotics strategy targets domestic deployment of 350,000 AI robots by 2030 and approximately 10 million by 2040 ([LINK](#)). The government also plans for ¥1tn of support over FY2026-2030 for domestic Physical AI foundation-model development ([LINK](#)), alongside targeted funding for advanced robot hardware. We view these initiatives as potentially lowering development barriers and encouraging private investment, creating opportunities for both technology suppliers and corporate adopters.

Corporate Governance reform - owning Japan AI not just for the AI story

While almost all the companies in our AI HW/SM universe have Price-to-Book ratios of above 1.0x, a metric set by the TSE during the early stage of the corporate governance reform in 2023, we believe that the corporate governance reform story also applies to this subset of Japanese companies. As the TSE has reiterated that all companies regardless of the current PBR level should be conscious of share price and cost of capital and respond to its request ([LINK](#)).

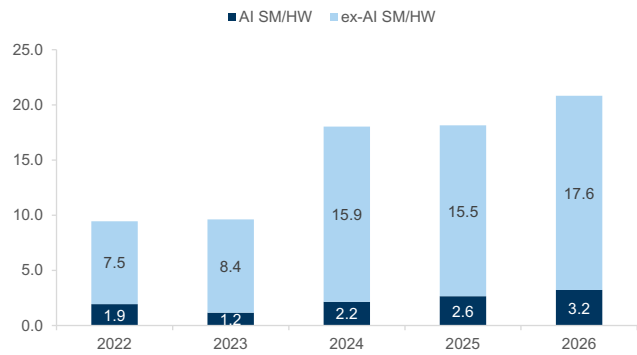
While our analysis suggests 30% of AI SM/HW have a NTM ROE that is below the market aggregate of 10.7% vs 56% for non-AI SM/HW, 58% of AI SM/HW companies have positive net cash vs 47% for non-AI SM/HW (ex-Financials). This suggests AI SM/HW companies are also under top-down TSE pressure and bottom-up investor engagement to improve. We believe the value unlock story is unique to Japan within the global AI stock universe and deserves investor attention.

Exhibit 45: Room for improvement in corporate governance even for AI SM/HW companies



Source: I/B/E/S, Toyo Keizai, FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 46: AI SM/HW companies are not overly committing on shareholder returns
Announced value of buyback



Source: Nikkei Value Search, Data compiled by Goldman Sachs Global Investment Research

Implementation ideas

We define our AI SM/HW universe of 123 Japanese stocks based on a 3-step process as detailed below. We have then used this approach to drive our stock screening for the **GS Japan AI semiconductor ecosystem basket (GSJPAISM)** and the **GS Japan AI data center hardware basket (GSJPAIHW)**.

1) Select AI-exposed sectors based on GICS industry classification

First, within the >¥50bn market cap and 6M ADTV > \$5mn universe, we remove stocks that are in GICS industries that are not exposed to AI semiconductor ecosystem (SM) and AI data-center hardware (HW) industries. The full list of excluded GICS industries is shown in [Exhibit 71](#) in the appendix.

2) Screening for companies with exposure to AI SM/HW-related RBICS sub-industries

From the sector-based results in step 1, we further filter for companies that have potential business/product level exposure to AI using RBICS sub-industry data, the full list of the 70 RBICS sub-industries selected is shown in [Exhibit 72](#) in the appendix. Companies with less than 5% sales exposure to these RBICS sub-industries are excluded.

FactSet Revere Business Industry Classification System (RBICS)

RBICS is a multi-sector classification framework designed to reveal a company's true sector exposures by analyzing its primary and ancillary revenue streams.

In essence, it normalizes non-standardized business segment reports, mapping companies' segment revenues to granular sectors. The RBICS matrix structure consists of 12 anchor sectors, 6 levels of depth, and over 1,800 individual sector groups. This results in a quantified multi-sector participation tree for each company, enabling easy comparison, screening, and grouping. This revenue-based approach makes RBICS an effective tool in providing high visibility into a company's true business operations, and enabling investors to screen for firms with specific revenue exposure to niche AI-related activities.

3) Textual analysis and GS bottom-up insights

GS Global Investment Research has published a wide range of in-depth reports on the topic of AI, encompassing semiconductors, hardware infrastructure, software, and AI-powered applications. For GS-covered stocks with available data, we only include stocks with AI-related sales or operating profit exposure of greater than 10% in FY25, or at least 5% in FY25 and are expected to expand the exposure to above 10% by FY28. For non GS-covered companies, we exclude companies with low AI exposure based on company filings and news search.

Based on the 3-step framework illustrated above, we construct the Japan AI semiconductor ecosystem universe and the GS Japan AI data center hardware universe shown in [Exhibit 66](#), [Exhibit 67](#), and [Exhibit 68](#) of the Appendix.

Based on stocks identified in our AI SM/HW universe, we further filter for stocks with higher liquidity and estimated AI-exposure to construct the **GS Japan AI semiconductor ecosystem basket (GSJPAISM)** and the **GS Japan AI data center hardware basket (GSJPAIHW)**. The criteria are as follows:

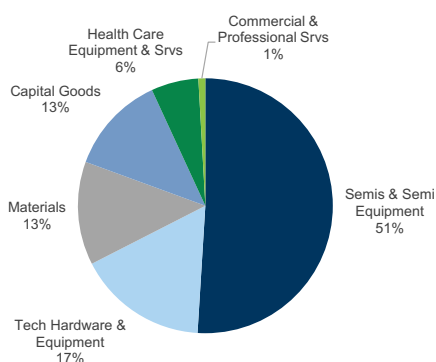
- Within our Japan AI semiconductor ecosystem stock universe, stocks with 6M ADTV > US\$20m.
- For GS-covered stocks with data availability, AI-related sales/OP exposure reaching at least 20% by FY28 based on Asia Technology 2H26 Investment Strategy report (July 7, 2026).
- For remaining stocks, sales exposure to selected RBICS sub-industries > 50% (Exhibit 72).

Japan AI semiconductor ecosystem basket (GSJPAISM)

The **Japan AI semiconductor value chain basket** contains five sub-layers, including (SM1) semiconductor materials and consumables, (SM2) wafer-fab equipment, metrology & critical subsystem/components, (SM3) advanced packaging, substrates, assembly & test, (SM4) AI-related semiconductor devices - memory, custom compute & edge AI, and (SM5) fab automation, material handling & utility infrastructure.

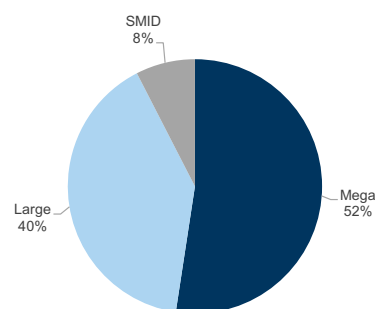
Weighting: Average market cap and 6M ADTV weighted, adjusted by relevancy multiplier based on estimated AI exposure. Estimated AI exposure for GS-covered stocks is based on sector analyst inputs if available, while remaining stocks use RBICS selected industries sales exposure multiplied by an adjustment factor of 0.3. Maximum/minimum stock weights are capped at 6%/0.5% respectively.

Exhibit 47: GICS industry group breakdown of GSJPAISM
Data as of 25 September 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 48: Size breakdown of GSJPAISM
Data as of 25 September 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 49: GS Japan AI semiconductor ecosystem basket (GSJPAISM)

Data as of 25 September 2026

GS Japan AI semiconductor ecosystem basket
(GSJPAISM)

					SM5: Fab automation, material handling & utility infrastructure									
Ticker	Name	GICS Industry Group	6M ADTV (\$mn)	Mcap (¥bn)	SM1	SM2	SM3	SM4	SM5	HW exposure	NTM PE	FY0 PB	Weight	
285A	KIOXIA HLDGS CORP	Semis & Semi Equipment	14,248	30,603				✓			4.6	21.8	6.0%	
6857	ADVANTEST CORP	Semis & Semi Equipment	1,742	24,888			✓				33.6	31.0	6.0%	
8035	TOKYO ELECTRON	Semis & Semi Equipment	1,412	26,453		✓	✓				28.0	12.6	6.0%	
4062	IBIDEN CO LTD	Tech Hardware & Equipment	1,013	6,621			✓				61.0	11.9	6.0%	
6920	LASERTEC CORP	Semis & Semi Equipment	947	3,857		✓					33.9	14.9	6.0%	
6146	DISCO CORPORATION	Semis & Semi Equipment	772	5,884			✓				28.2	10.0	6.0%	
6861	KEYENCE CORP	Tech Hardware & Equipment	349	19,359					✓		32.2	5.6	6.0%	
4063	SHIN-ETSU CHEMICAL	Materials	321	11,690	✓						18.4	2.5	6.0%	
7741	HOYA CORP	Health Care Equipment & Svcs	177	7,864	✓						26.9	7.7	6.0%	
6525	KOKUSAI ELECTRIC C	Semis & Semi Equipment	413	2,209		✓					30.9	9.9	4.6%	
6723	RENESAS ELECTRONIC	Semis & Semi Equipment	387	6,506				✓		✓	13.6	2.6	4.4%	
6273	SMC CORP	Capital Goods	149	4,447		✓			✓		19.9	2.1	3.9%	
7735	SCREEN HOLDINGS CO	Semis & Semi Equipment	244	2,621		✓					19.3	5.3	3.1%	
5016	JX ADVANCED METALS	Materials	671	3,374	✓					✓	18.4	4.5	2.4%	
3110	NITTO BOSEKI CO	Capital Goods	225	562			✓			✓	22.5	3.1	1.9%	
3436	SUMCO CORPORATION	Semis & Semi Equipment	351	1,091	✓						55.8	1.9	1.7%	
4004	RESONAC HOLDINGS C	Materials	298	2,860	✓		✓			✓	19.6	3.9	1.5%	
5333	NGK CORP	Capital Goods	64	1,549		✓					16.5	1.9	1.4%	
6361	EBARA CORP	Capital Goods	116	2,031		✓				✓	17.8	4.0	1.4%	
5706	mitsui kinzoku co	Materials	448	1,319			✓			✓	13.7	3.2	1.3%	
4186	TOKYO OHKA KOGYO	Materials	60	1,068	✓		✓				20.6	4.4	1.1%	
7729	TOKYO SEIMITSU CO	Semis & Semi Equipment	48	747			✓				19.4	3.8	1.0%	
6526	SOCIONEXT INC	Semis & Semi Equipment	134	404				✓			23.9	3.0	1.0%	
6871	MICRONICS JAPAN CO	Semis & Semi Equipment	111	554			✓				18.8	8.1	1.0%	
6787	MEIKO ELECTRONICS	Tech Hardware & Equipment	91	640			✓			✓	18.4	4.7	0.9%	
6856	HORIBA LTD	Tech Hardware & Equipment	33	1,007		✓					16.4	2.9	0.9%	
7911	TOPPAN HOLDINGS IN	Commercial & Professional Svcs	55	1,449			✓				16.3	1.0	0.9%	
6645	OMRON CORP	Tech Hardware & Equipment	58	1,304					✓		21.0	1.5	0.8%	
5344	MARUWA CO LTD	Tech Hardware & Equipment	64	688		✓				✓	23.8	4.7	0.8%	
4182	mitsubishi gas chm	Materials	41	827	✓		✓				13.2	1.2	0.8%	
6481	THK CO LTD	Capital Goods	49	766					✓		16.4	2.8	0.8%	
6323	RORZE CORP	Semis & Semi Equipment	57	668					✓		18.0	5.0	0.8%	
6368	ORGANO CORP	Capital Goods	26	588					✓		17.5	4.1	0.6%	
268A	RIGAKU HOLDINGS CO	Tech Hardware & Equipment	52	378		✓					23.6	4.3	0.5%	
6754	ANRITSU CORP	Tech Hardware & Equipment	40	462			✓			✓	23.8	3.3	0.5%	
6370	KURITA WATER INDS	Capital Goods	30	897					✓		16.7	2.5	0.5%	
6315	TOWA CORP	Semis & Semi Equipment	78	165			✓				16.6	2.3	0.5%	
6590	SHIBAURA MECHATRON	Semis & Semi Equipment	62	264		✓	✓				14.1	4.5	0.5%	
6227	AIMECHATEC LTD	Semis & Semi Equipment	57	125			✓				22.2	8.6	0.5%	
5805	SWCC CORPORATION	Capital Goods	40	417			✓			✓	16.3	4.1	0.5%	
6855	JAPAN ELECTRON MAT	Semis & Semi Equipment	34	111			✓				14.8	2.5	0.5%	
6728	ULVAC INC	Semis & Semi Equipment	32	376		✓					16.9	1.6	0.5%	
6407	CKD CORP	Capital Goods	29	358		✓					15.2	2.3	0.5%	
429A	TEKSCEND PHOTOMASK	Semis & Semi Equipment	29	386	✓						14.1	2.2	0.5%	
6622	DAIHEN CORP	Capital Goods	23	322		✓			✓	✓	14.7	2.0	0.5%	
6254	NOMURA MICRO SCIEN	Capital Goods	22	139					✓		10.6	3.3	0.5%	
6941	YAMAICHI ELECTRONI	Semis & Semi Equipment	22	176			✓			✓	13.4	3.4	0.5%	
indicate GS covered stocks		Median	64	897								18	4	

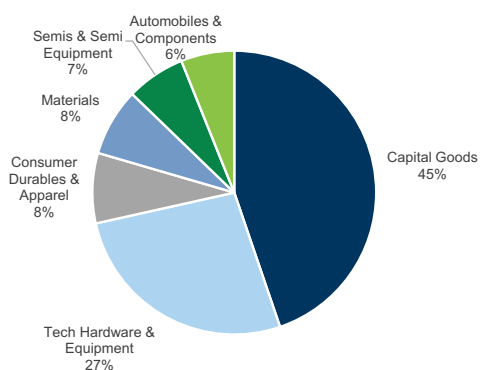
Source: FactSet, Goldman Sachs Global Investment Research

Japan AI infrastructure hardware basket (GSJPAIHW)

The AI infrastructure hardware basket contains 3 sub-themes, including (HW1) High-speed interconnect & data transmission, (HW2) Server, PCB, rack power & cooling components, and (HW3) Data-center power & thermal hardware.

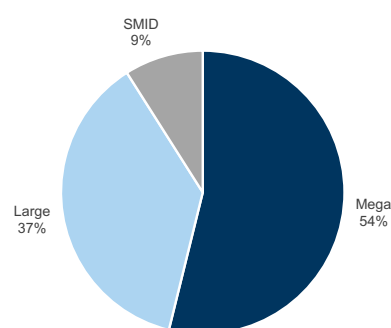
Weighting: Average market cap and 6M ADTV weighted, adjusted by relevancy multiplier based on estimated AI exposure. Estimated AI exposure for GS-covered stocks is based on sector analyst inputs if available, while remaining stocks use RBICS selected industries sales exposure multiplied by an adjustment factor of 0.3. Maximum/minimum stock weights are capped at 8%/1% respectively.

Exhibit 50: GICS industry group breakdown of GSJPAIHW
Data as of 25 September, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 51: Size breakdown of GSJPAIHW
Data as of 25 September, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 52: GS Japan AI data center hardware basket (GSJPAIHW)

Data as of 25 September 2026

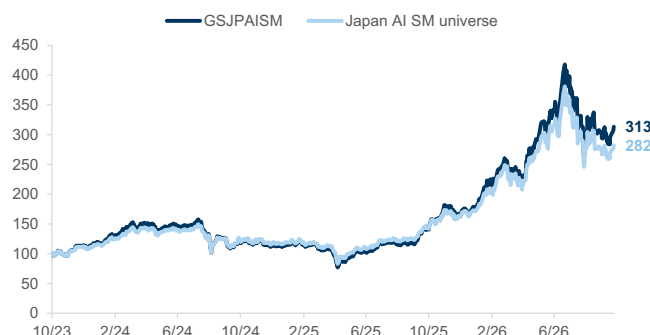
GS Japan AI data center hardware basket
(GSJPAIHW)

 GS Japan AI data center hardware basket (GSJPAIHW)					HW1: High-speed interconnect & data transmission						
					HW2: Server, PCB, rack power & cooling components						
					HW3: Data center power & thermal hardware						
Ticker	Name	GICS Industry Group	6M ADTV (\$mn)	Mcap (¥bn)	HW1	HW2	HW3	SM exposure	NTM PE	FY0 PB	Weight
5803	FUJIKURA	Capital Goods	1,803	8,851	✓	✓			25.5	14.7	8.0%
6981	MURATA MFG CO	Tech Hardware & Equipment	1,373	15,747		✓			33.4	5.4	8.0%
6501	HITACHI	Capital Goods	403	25,109			✓		23.0	3.8	8.0%
6752	PANASONIC HLDGS CO	Consumer Durables & Apparel	269	11,262		✓	✓		19.9	2.1	8.0%
5801	FURUKAWA ELECTRIC	Capital Goods	1,010	2,704	✓	✓			21.7	6.5	7.4%
6503	mitsubishi elec CP	Capital Goods	232	11,071	✓		✓		18.9	2.4	7.2%
6976	TAIYO YUDEN CO LTD	Tech Hardware & Equipment	1,599	1,260		✓			30.1	3.3	6.3%
5802	SUMITOMO ELECTRIC	Automobiles & Components	406	7,022	✓				17.8	2.5	6.1%
6723	RENESAS ELECTRONIC	Semis & Semi Equipment	387	6,506		✓		✓	13.6	2.6	5.7%
5016	JX ADVANCED METALS	Materials	671	3,374	✓	✓		✓	18.4	4.5	3.6%
6594	NIDEC CORPORATION	Capital Goods	96	3,387		✓	✓		16.4	1.9	3.5%
3110	NITTO BOSEKI CO	Capital Goods	225	562		✓		✓	22.5	3.1	3.1%
6762	TDK CORP	Tech Hardware & Equipment	272	5,789		✓			20.3	2.6	2.9%
5706	mitsui kinzoku co	Materials	448	1,319	✓	✓		✓	13.7	3.2	2.1%
4004	RESONAC HOLDINGS C	Materials	298	2,860		✓		✓	19.6	3.9	2.0%
6479	MINEBEA MITSUMI	Capital Goods	77	1,535		✓			15.2	1.6	1.9%
6361	EBARA CORP	Capital Goods	116	2,031			✓	✓	17.8	4.0	1.7%
6787	MEIKO ELECTRONICS	Tech Hardware & Equipment	91	640		✓		✓	18.4	4.7	1.3%
5344	MARUWA CO LTD	Tech Hardware & Equipment	64	688		✓		✓	23.8	4.7	1.1%
6965	HAMAMATSU PHOTONIC	Tech Hardware & Equipment	58	748	✓				33.3	2.2	1.1%
4980	DEXERIALS CORP	Tech Hardware & Equipment	49	538	✓				16.5	4.7	1.0%
6834	SEIKOH GIKEN CO	Tech Hardware & Equipment	43	284	✓				26.4	8.0	1.0%
5805	SWCC CORPORATION	Capital Goods	40	417	✓			✓	16.3	4.1	1.0%
6754	ANRITSU CORP	Tech Hardware & Equipment	40	462	✓			✓	23.8	3.3	1.0%
6134	FUJI CORPORATION	Capital Goods	39	729		✓			13.4	2.8	1.0%
6278	UNION TOOL CO	Capital Goods	38	283		✓			18.1	3.1	1.0%
6779	NIHON DEMPY KOGYO	Tech Hardware & Equipment	31	75	✓				21.7	2.3	1.0%
6997	NIPPON CHEMI-CON	Tech Hardware & Equipment	29	73		✓			12.7	1.3	1.0%
6777	SANTEC HLDGS CORP	Tech Hardware & Equipment	29	262	✓				23.9	9.2	1.0%
6622	DAIHEN CORP	Capital Goods	23	322			✓	✓	14.7	2.0	1.0%
6941	YAMAICHI ELECTRONI	Semis & Semi Equipment	22	176	✓			✓	13.4	3.4	1.0%
indicate GS covered stocks		Median	96	1,260					18.9	3.3	

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 53: Performance of GSJPAISM vs broader Japan AI SM universe

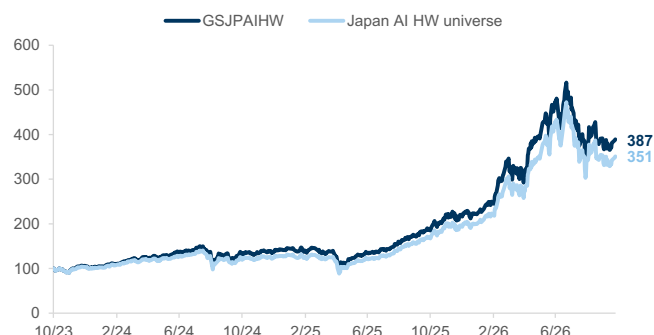
Data as of Sep 25, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 54: Performance of GSJPAIHW vs broader Japan AI HW universe

Data as of Sep 25, 2026



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 55: Summary of GSJPAISM and GSJPAIHW baskets

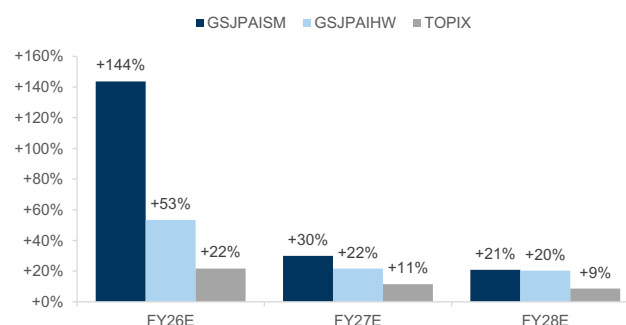
Data as of Sep 25, 2026

		GSJPAISM	GSJPAIHW
Total liquidity	# of stocks	47	31
Total Listed Market Cap (¥bn)		181,076	116,087
Total 6M ADTV (\$mn)		25,725	10,282
Valuations (Aggregate)	FY27E PE	14.2	17.9
	FY26E PB	4.4	2.8
	Fwd 2Y PEG	0.6	0.4
	Fwd 3Y PEG	0.8	0.3
	fwd-24m PE z-score (1Y)	-1.7	-0.6
	fwd-24m PE z-score (3Y)	-1.5	0.5
EPS Growth (%)	FY26E	144%	53%
	FY27E	30%	22%
	FY28E	21%	20%
Price return	2024	-4%	38%
	2025	54%	53%
	2026 (YTD)	78%	76%

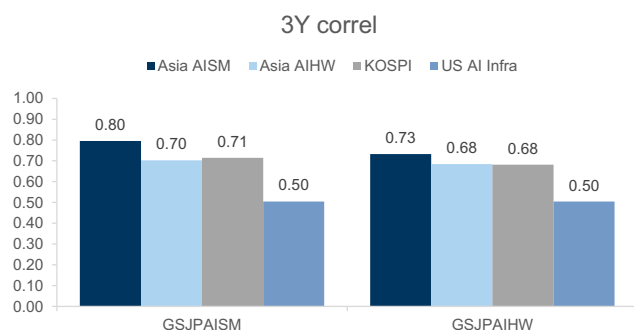
Source: I/B/E/S, FactSet, Goldman Sachs Global Investment Research

Exhibit 56: GSJPAISM and GSJPAIHW EPS growth estimate vs TOPIX

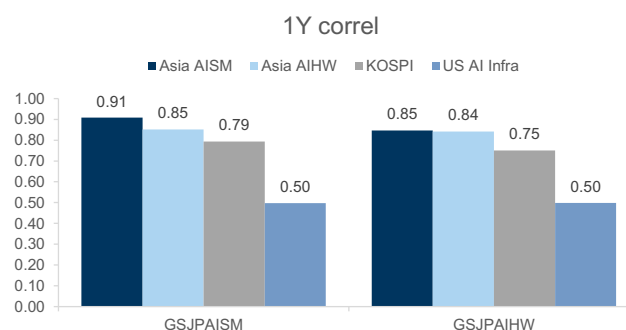
Based on consensus estimates



Source: I/B/E/S, FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 57: GSJPAISM and GSJPAIHW 3 year correlation with Asia AI SM/HW baskets, KOSPI, and US AI Phase 2 basket


Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 58: GSJPAISM and GSJPAIHW 1 year correlation with Asia AI SM/HW baskets, KOSPI, and US AI Phase 2 basket


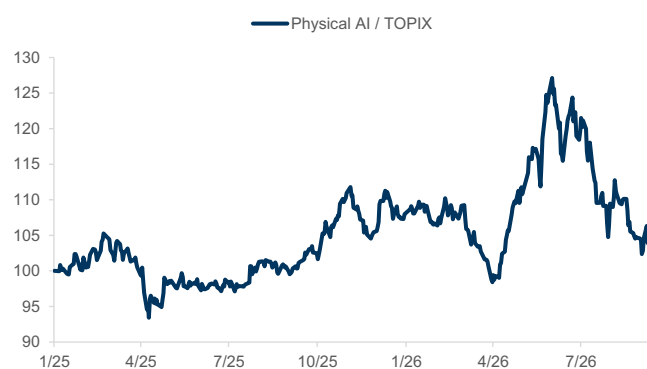
Source: FactSet, Goldman Sachs Global Investment Research

Japan Physical AI screen

Based on our prior work on the thematic in [Five investment themes for equity investors in 2026](#) (5 January 2026), ([JP link](#)) we fine-tune our physical AI screening to reflect the latest strategic partnerships and sector analyst views. The criteria are as follows:

- Market capitalization of greater than ¥50bn
- 6M ADTV > US\$5m
- Strategic investments, partnerships, product tech initiatives related to Physical AI; or
- Stocks highlighted by our GS analyst in [Humanoids & Auto: Building Physical AI Together, Framing the Forward Progress of Humanoids \(with a Studied Focus on the Logistics Landscape\)](#)

Exhibit 59: Equal-weighted performance of Japan Physical AI screen relative to TOPIX



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 60: Japan Physical AI screen

Data as of 18 September 2026

Ticker	Name	Sector	6M ADTV (\$ mn)	Mcap (¥ bn)	PBR FY0 (x)	PER NTM (x)	ROE FY26E (%)	FY27E (%)	Div Yield FY26E (%)	FY27E (%)
9984	SOFTBANK GROUP CO	Information & Communication	2,414	36,070	2.1	47.2	4	3	0.2	0.2
6981	MURATA MFG CO	Electric Appliances	1,328	15,239	5.2	32.6	13	17	0.9	1.1
7203	TOYOTA MOTOR CORP	Transportation Equipment	510	44,150	1.0	9.3	9	10	3.4	3.8
6758	SONY GROUP CORPORA	Electric Appliances	420	22,054	2.7	16.5	15	14	0.9	1.0
6501	HITACHI	Electric Appliances	400	25,054	3.8	23.0	15	17	1.1	1.2
6723	RENESAS ELECTRONIC	Electric Appliances	379	6,190	2.5	12.9	19	14	1.0	1.1
6954	FANUC CORP	Electric Appliances	280	5,748	2.9	25.7	11	12	2.3	2.5
7012	KAWASAKI HEAVY IND	Transportation Equipment	241	2,109	2.3	15.6	13	13	1.5	1.8
6503	mitsubishi elec CP	Electric Appliances	229	10,832	2.3	18.5	12	12	1.3	1.4
6701	NEC CORP	Electric Appliances	208	6,565	2.9	19.8	13	14	0.9	1.0
6506	YASKAWA ELEC CORP	Electric Appliances	196	1,169	2.4	21.4	9	11	1.7	1.9
6702	FUJITSU	Electric Appliances	185	6,923	3.4	20.6	15	17	1.3	1.5
7267	HONDA MOTOR CO	Transportation Equipment	169	7,656	0.6	10.1	4	7	4.2	4.4
6479	MINEBEA MITSUMI	Electric Appliances	76	1,476	1.5	14.6	10	10	1.7	1.9
6383	DAIFUKU CO LTD	Machinery	74	2,070	4.4	20.3	20	19	1.6	1.8
6324	HARMONIC DRIVE SYS	Machinery	68	597	7.3	72.6	8	11	0.4	0.6
6645	OMRON CORP	Electric Appliances	58	1,272	1.5	20.4	6	7	1.9	2.0
7259	aisin CORPORATION	Transportation Equipment	36	1,622	0.7	9.3	7	8	3.4	4.0
7211	mitsubishi motor C	Transportation Equipment	29	538	0.5	12.8	3	5	2.7	3.1
6302	SUMITOMO HEAVY IND	Machinery	28	735	1.1	14.1	6	7	2.4	2.6
6471	NSK LTD	Machinery	18	541	0.8	14.6	5	6	3.3	3.5
6473	JTEKT CORPORATION	Machinery	16	664	0.8	11.2	6	8	3.3	4.0
Average			335	9,058	2.4	21.1	10	11	1.9	2.1
Median			190	3,928	2.3	17.5	10	11	1.6	1.8

Source: I/B/E/S, Toyo Keizai, FactSet, Data compiled by Goldman Sachs Global Investment Research

Stocks with negative correlation to Japan AI universe

While we remain constructive on the AI theme, it is also useful to identify stocks most negatively correlated with our AI-universe for when positioning gets extended. We screen for stocks with negative correlation with Japan AI SM/HW performance over the last 3 years. The screening criteria are:

- Market capitalization of greater than ¥50bn
- 6M ADTV > \$5m
- Stocks with negative correlation with Japan AI SM/HW stock universe.

This resulted in a screen with 51 names, sorted by sector.

Exhibit 61: Stocks that are negatively correlated with Japan AI stock universe

Data as of 18 September 2026

vs SM/HW universe												
Ticker	Name	Sector	3Y Correl	3Y Beta	6M ADTV (\$ mn)	Mcap (¥ bn)	PBR FY0 (x)	PER NTM (x)	ROE FY26E (%)	FY27E (%)	Div Yield FY26E (%)	FY27E (%)
4452	KAO CORP	Chemicals	(0.03)	(0.02)	68	3,155	3.0	21.4	13	13	2.2	2.3
2269	MEIJI HOLDINGS CO	Foods	(0.20)	(0.11)	26	1,229	1.5	17.2	8	9	2.6	2.7
2282	NH FOODS LTD	Foods	(0.13)	(0.09)	22	573	1.1	13.8	7	8	2.9	3.1
2871	NICHIREI CORP	Foods	(0.13)	(0.08)	19	568	1.9	17.8	9	10	2.3	2.4
2809	KEWPIE CORP	Foods	(0.10)	(0.07)	11	628	1.9	21.3	8	9	1.6	1.7
2267	YAKULT HONSHA CO	Foods	(0.09)	(0.06)	25	852	1.4	18.2	7	7	2.6	2.6
2201	MORINAGA & CO	Foods	(0.06)	(0.03)	5	214	1.5	12.2	11	11	2.8	3.1
2212	YAMAZAKI BAKING	Foods	(0.05)	(0.04)	14	803	1.6	15.8	9	9	1.7	1.8
2593	ITO EN LTD	Foods	(0.04)	(0.03)	9	270	2.1	26.1	8	9	1.6	1.8
2810	HOUSE FOODS GP INC	Foods	(0.02)	(0.01)	6	366	1.2	20.7	5	6	2.5	2.6
2206	EZAKI GLICO CO	Foods	(0.02)	(0.01)	8	345	1.2	29.0	3	4	1.9	2.0
2264	MORINAGA MILK IND	Foods	(0.01)	(0.01)	15	394	1.3	14.3	9	9	2.4	2.8
9602	TOHO CO LTD	Information & Communication	(0.25)	(0.21)	36	1,343	2.6	25.1	10	10	1.4	1.6
4722	FUTURE CORPORATION	Information & Communication	(0.08)	(0.07)	9	231	3.4	14.6	19	20	2.0	2.4
9433	KDDI CORP	Information & Communication	(0.08)	(0.04)	159	12,263	2.3	14.9	14	14	2.7	2.9
4816	TOEI ANIMATION	Information & Communication	(0.03)	(0.03)	7	629	3.6	23.7	14	14	1.6	1.8
9142	KYUSHU RAILWAY COR	Land Transportation	(0.14)	(0.07)	15	535	1.1	10.0	10	10	3.5	3.7
9020	EAST JAPAN RAILWAY	Land Transportation	(0.14)	(0.08)	73	3,925	1.3	13.8	9	9	2.5	3.0
9006	KEIKYU CORPORATION	Land Transportation	(0.10)	(0.05)	8	432	1.1	13.8	8	7	2.9	2.9
9143	SG HOLDINGS CO LTD	Land Transportation	(0.08)	(0.06)	14	957	1.6	13.4	11	12	3.6	3.7
9005	TOKYU CORP	Land Transportation	(0.07)	(0.04)	33	1,039	1.0	10.2	10	9	1.9	2.1
9022	CENTRAL JAPAN RLWY	Land Transportation	(0.05)	(0.04)	67	3,797	0.7	7.3	9	9	0.8	0.9
9042	HANKYU HANSHIN HLD	Land Transportation	(0.04)	(0.03)	32	1,160	1.0	12.9	7	8	2.2	2.3
9003	SOTETSU HOLDINGS	Land Transportation	(0.02)	(0.01)	5	259	1.2	10.3	12	12	2.8	3.0
9024	SEIBU HOLDINGS INC	Land Transportation	(0.01)	(0.02)	40	1,105	1.6	22.9	6	8	1.2	1.3
9007	ODAKYU ELEC RLWY	Land Transportation	(0.01)	(0.01)	15	663	1.2	15.0	8	8	3.3	3.5
9076	SEINO HOLDINGS	Land Transportation	(0.00)	(0.00)	6	513	1.0	14.8	6	7	3.9	4.0
1878	DAITO TRUST CONST	Real Estate	(0.00)	(0.00)	34	1,155	2.2	10.2	21	19	5.0	5.0
3349	COSMOS PHARMACEUTI	Retail Trade	(0.29)	(0.25)	17	515	1.8	15.6	11	11	1.3	1.4
8267	AEON CO LTD	Retail Trade	(0.19)	(0.18)	87	3,656	3.0	44.1	6	8	1.1	1.2
2726	PAL GROUP HOLDINGS	Retail Trade	(0.18)	(0.23)	9	275	3.1	13.2	21	21	2.7	3.0
7581	SAIZERIYA COMPANY	Retail Trade	(0.15)	(0.18)	25	335	2.7	21.0	10	11	0.5	0.6
2782	SERIA CO LTD	Retail Trade	(0.13)	(0.12)	8	359	3.2	18.7	16	13	1.7	1.8
9843	NITORI HOLDINGS	Retail Trade	(0.13)	(0.14)	63	1,734	1.7	18.5	9	9	1.1	1.1
9861	YOSHINOYA HOLDINGS	Retail Trade	(0.12)	(0.08)	11	237	3.5	38.3	8	9	0.7	0.7
8252	MARUI GROUP	Retail Trade	(0.10)	(0.05)	18	529	2.1	16.1	12	13	4.7	4.9
3197	SKYLARK HOLDINGS C	Retail Trade	(0.10)	(0.08)	31	742	4.0	32.9	11	11	0.8	0.9
3391	TSURUHA HOLDINGS I	Retail Trade	(0.09)	(0.08)	28	983	1.1	18.8	6	7	2.3	2.5
3387	CREATE RESTAURANTS	Retail Trade	(0.09)	(0.06)	7	347	7.3	52.9	12	14	0.7	0.7
2702	MCDONALD'S HOLDING	Retail Trade	(0.08)	(0.04)	31	1,045	3.7	27.6	13	12	0.8	0.9
3397	TORIDOLL HLDGS COR	Retail Trade	(0.07)	(0.05)	9	395	4.2	45.5	8	10	0.3	0.4
3549	KUSURI NO AOKI HLD	Retail Trade	(0.07)	(0.06)	6	382	2.7	19.3	13	14	1.6	1.7
9956	VALOR HOLDINGS	Retail Trade	(0.06)	(0.04)	6	201	1.0	10.0	9	9	2.3	2.4
2730	EDION CORP	Retail Trade	(0.05)	(0.03)	5	282	1.1	15.6	7	7	2.0	2.0
2670	ABC-MART INC	Retail Trade	(0.03)	(0.02)	12	665	1.7	13.7	12	12	3.0	3.1
8227	SHIMAMURA CO	Retail Trade	(0.03)	(0.02)	16	705	1.4	13.4	10	9	2.5	2.7
4784	GMO INTERNET INC	Services	(0.01)	(0.02)	7	165	10.8	24.4	45	51	4.1	2.8
9735	SECOM CO	Services	(0.00)	(0.00)	53	3,003	2.0	20.8	9	10	1.8	2.1
2331	ALSOK CO LTD	Services	(0.00)	(0.00)	10	628	1.6	14.7	10	10	2.9	3.1
3038	KOBE BUSSAN CO LTD	Wholesale Trade	(0.36)	(0.28)	33	831	4.3	20.1	19	18	1.1	1.1
7459	MEDIPAL HLDG CORP	Wholesale Trade	(0.08)	(0.05)	8	628	0.9	13.7	7	7	2.5	2.8
	Average				25	1,138	2.3	19.4	10.8	11.3	2.2	2.3
	Median				15	628	1.7	16.1	9.4	9.6	2.2	2.4

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

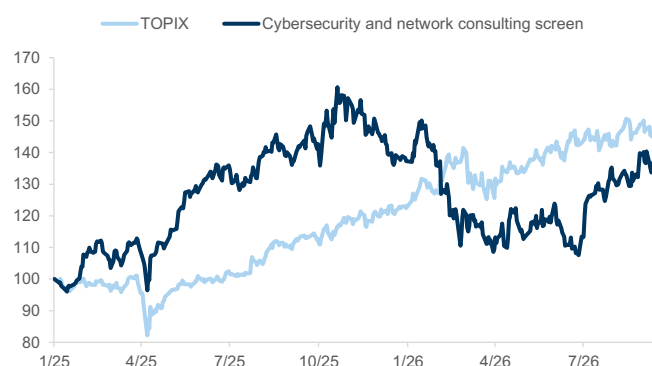
Japan cybersecurity screen

To address investor concerns around AI-related cyber attacks, we screen for Japanese stocks likely to benefit from increased demand for cybersecurity-related IT services. The equal-weighted performance of the screen is shown in [Exhibit 62](#), the screen corrected sharply at the start of the year amid concerns around AI-disruption risks. The screen subsequently recovered from July but remains below last year's peak. The screening criteria are as follows:

- Market capitalization of greater than ¥50bn
- 6M ADTV > US\$5m
- Companies with RBICS sub-industry sales exposure to (1) network security software, (2) general infrastructure and network consulting, or (3) design, integration and implementation consulting

Exhibit 62: Performance of cybersecurity screen

Equal-weighted



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 63: Japan cyber-security screen

Data as of Sep 14, 2026

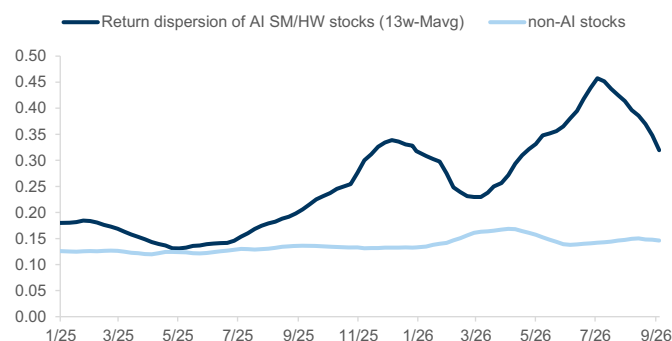
Ticker	Name	GICS Industry	6M ADTV (\$ mn)	Mcap (\$ bn)	PBR FY0 (x)	PER NTM (x)	ROE		Div Yield	
							FY26E (%)	FY27E (%)	FY26E (%)	FY27E (%)
6701	NEC CORP	IT Services	206.8	6,615	2.9	20.1	13	14	0.9	1.0
4307	NOMURA RESEARCH IN	IT Services	96.5	2,922	6.9	22.5	28	28	1.7	1.8
6532	BAYCURRENT INC	Professional Services	60.8	1,155	10.0	20.8	39	42	1.7	2.1
4704	TREND MICRO INC	Software	40.2	855	6.2	21.4	26	28	3.0	3.4
4768	OTSUKA CORP	IT Services	32.2	1,375	3.5	19.3	16	16	2.8	3.0
8056	BIPROGY INC	IT Services	13.8	481	2.6	13.8	17	17	2.9	3.2
3774	INTERNET INITIATIV	Diversified Telecom Services	12.3	633	3.9	22.3	15	16	1.3	1.4
4812	DENTSU SOKEN INC	IT Services	9.9	560	5.6	27.7	18	18	1.6	1.8
4722	FUTURE CORPORATION	IT Services	8.7	231	3.4	14.6	19	20	2.0	2.4
4373	SIMPLEX HLDGS INC	IT Services	6.9	275	5.0	18.6	23	25	2.0	2.4
2327	NS SOLUTIONS CORP	IT Services	5.9	725	2.6	21.4	11	12	2.3	2.4
3692	FFRI SECURITY INC	Software	5.4	57	14.6	73.7	34	41	0.3	0.3
4413	BAUDROIE INC	IT Services	5.3	95	12.0	24.5	33	34	0.3	0.5
Average			38.8	1,229	6.1	24.7	23	24	1.7	2.0
Median			12.3	633	5.0	21.4	19	20	1.7	2.1

Source: I/B/E/S, FactSet, Data compiled by Goldman Sachs Global Investment Research

GS Buy-rated stocks in GS Japan AI SM/HW universe

Within our Japan AI SM and HW universe, we also screen for GS-covered Buy-rated names. This provides a screen of 42 names.

Exhibit 64: Return dispersion of AI universe (SM/HW) has declined, but still much higher than non-AI stocks, which suggests the right stock selection is rewarded more



Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

Exhibit 65: GS Buy-rated stocks in Japan AI SM and HW universe

Data as of 25 September 2026

Ticker	Name	GICS Industry Group	AISM	AIHW	6M ADTV (\$ mn)	Mcap (¥ bn)	PBR FY0 (x)	PER NTM (x)	ROE FY26E (%)	ROE FY27E (%)	Div Yield FY26E (%)	Div Yield FY27E (%)	GS Rating
285A	KIOXIA HLDGS CORP	Semis & Semi Equipment	✓		14,248	30,603	21.8	4.6	152	85	1.4	3.2	Buy
8035	TOKYO ELECTRON	Semis & Semi Equipment	✓		1,412	26,453	12.6	28.0	35	41	1.5	2.0	Buy
6501	HITACHI	Capital Goods		✓	403	25,109	3.8	23.0	15	17	1.1	1.2	Buy
6861	KEYENCE CORP	Tech Hardware & Equipment	✓		349	19,359	5.6	32.2	15	16	0.8	0.9	Buy
6981	MURATA MFG CO	Tech Hardware & Equipment		✓	1,373	15,747	5.4	33.4	13	17	0.9	1.0	Buy
4063	SHIN-ETSU CHEMICAL	Materials	✓		321	11,690	2.5	18.4	12	14	2.0	2.3	Buy
6752	PANASONIC HLDGS CO	Consumer Durables & Apparel		✓	269	11,262	2.1	19.9	9	10	1.2	1.5	Buy
6503	MITSUBISHI ELEC CP	Capital Goods		✓	232	11,071	2.4	18.9	12	12	1.3	1.4	Buy
5803	FUJIKURA	Capital Goods		✓	1,803	8,851	14.7	25.5	47	42	1.4	1.7	Buy
7741	HOYA CORP	Health Care Equipment & Services	✓		177	7,864	7.7	26.9	26	29	1.4	1.6	Buy
5802	SUMITOMO ELECTRIC	Automobiles & Components		✓	406	7,022	2.5	17.8	12	13	1.9	2.3	Buy
4062	IBIDEN CO LTD	Tech Hardware & Equipment	✓		1,013	6,621	11.9	61.0	14	18	0.2	0.2	Buy
6723	RENEAS ELECTRONIC	Semis & Semi Equipment	✓	✓	387	6,506	2.6	13.6	19	14	0.9	1.0	Buy
6146	DISCO CORPORATION	Semis & Semi Equipment	✓		772	5,884	10.0	28.2	28	29	1.2	1.5	Buy
6762	TDK CORP	Tech Hardware & Equipment		✓	272	5,789	2.6	20.3	11	12	1.4	1.6	Buy
2802	AJINOMOTO CO INC	Food Beverage & Tobacco	✓		143	4,939	6.3	31.2	18	21	1.1	1.3	Buy
6920	LASERTEC CORP	Semis & Semi Equipment	✓		947	3,857	14.9	33.9	29	35	0.8	0.9	Buy
5016	JX ADVANCED METALS	Materials	✓	✓	671	3,374	4.5	18.4	24	23	0.9	1.3	Buy
4004	RESONAC HOLDINGS C	Materials	✓	✓	298	2,860	3.9	19.6	14	18	0.4	0.5	Buy
5801	FURUKAWA ELECTRIC	Capital Goods		✓	1,010	2,704	6.5	21.7	24	24	0.6	0.7	Buy
3407	ASAHI KASEI CORP	Materials	✓		56	2,247	1.1	12.4	8	9	2.8	3.0	Buy
6383	DAIFUKU CO LTD	Capital Goods	✓		74	2,108	4.5	20.7	20	19	1.6	1.8	Buy
6361	EBARA CORP	Capital Goods	✓	✓	116	2,031	4.0	17.8	18	19	1.7	2.0	Buy
6963	ROHM CO LTD	Semis & Semi Equipment		✓	146	1,935	2.4	36.3	5	7	1.0	1.0	Buy
6479	MINEBEA MITSUMI	Capital Goods		✓	77	1,535	1.6	15.2	10	10	1.6	1.9	Buy
7911	TOPPAN HOLDINGS IN	Commercial & Professional Services	✓		55	1,449	1.0	16.3	6	7	1.2	1.3	Buy
5706	mitsui KINZOKU CO	Materials	✓	✓	448	1,319	3.2	13.7	20	19	1.3	1.4	Buy
6645	OMRON CORP	Tech Hardware & Equipment	✓		58	1,304	1.5	21.0	6	7	1.9	2.0	Buy
6976	TAIYO YUDEN CO LTD	Tech Hardware & Equipment		✓	1,599	1,260	3.3	30.1	8	14	1.0	1.2	Buy
6506	YASKAWA ELEC CORP	Capital Goods	✓		198	1,195	2.4	22.1	9	11	1.6	1.8	Buy
3436	SUMCO CORPORATION	Semis & Semi Equipment	✓		351	1,091	1.9	55.8	(2)	5	0.6	0.9	Buy
4186	TOKYO OHKA KOGYO	Materials	✓		60	1,068	4.4	20.6	18	19	1.0	1.2	Buy
5332	TOTO LTD	Capital Goods	✓		61	984	1.8	18.3	9	10	2.0	2.6	Buy
6370	KURITA WATER INDS	Capital Goods	✓		30	897	2.5	16.7	14	15	1.8	2.0	Buy
5344	MARUWA CO LTD	Tech Hardware & Equipment	✓	✓	64	688	4.7	23.8	16	16	0.2	0.3	Buy
6368	ORGANO CORP	Capital Goods	✓		26	588	4.1	17.5	20	20	1.8	2.0	Buy
3110	NITTO BOSEKI CO	Capital Goods	✓	✓	225	562	3.1	22.5	12	14	1.1	1.4	Buy
6674	GS YUASA CORP	Automobiles & Components		✓	36	542	1.4	12.5	10	10	1.9	2.2	Buy
6754	ANRITSU CORP	Tech Hardware & Equipment	✓	✓	40	462	3.3	23.8	12	14	1.6	1.9	Buy
5805	SWCC CORPORATION	Capital Goods	✓	✓	40	417	4.1	16.3	22	22	2.2	2.5	Buy
6407	CKD CORP	Capital Goods	✓		29	358	2.3	15.2	13	14	2.3	2.7	Buy
6254	NOMURA MICRO SCIEN	Capital Goods	✓		22	139	3.3	10.6	25	26	2.5	2.8	Buy
Average					722	5,756	4.9	22.8	19.3	19.0	1.4	1.6	
Median					229	2,178	3.3	20.5	14.2	15.8	1.3	1.5	

Source: Goldman Sachs Global Investment Research

Appendix

Exhibit 66: Full Japan AI semiconductor ecosystem universe

Data as of Sep 25, 2026



Japan AI semiconductor ecosystem universe

					SM1: Semiconductor materials and consumables								
					SM2: Wafer-fab equipment, metrology & critical subsystem/components					SM3: Advanced packaging, substrates, assembly & test			
										SM4: AI-related semiconductor devices - memory, compute & edge AI			
										SM5: Fab automation, material handling & utility infrastructure			
Ticker	Name	GICS Industry Group	6M ADTV (\$mn)	Mcap (¥bn)	SM1	SM2	SM3	SM4	SM5	HW exposure	NTM PE	FY0 PB	In SM basket
285A	KIOXIA HLDGS CORP	Semis & Semi Equipment	14,248	30,603				✓			4.6	21.8	✓
6857	ADVANTEST CORP	Semis & Semi Equipment	1,742	24,888			✓				33.6	31.0	✓
8035	TOKYO ELECTRON	Semis & Semi Equipment	1,412	26,453		✓	✓				28.0	12.6	✓
4062	IBIDEN CO LTD	Tech Hardware & Equipment	1,013	6,621			✓				61.0	11.9	✓
6920	LASERTEC CORP	Semis & Semi Equipment	947	3,857		✓					33.9	14.9	✓
6146	DISCO CORPORATION	Semis & Semi Equipment	772	5,884			✓				28.2	10.0	✓
5016	JX ADVANCED METALS	Materials	671	3,374	✓					✓	18.4	4.5	✓
5706	MITSUI KINZOKU CO	Materials	448	1,319			✓			✓	13.7	3.2	✓
6525	KOKUSAI ELECTRIC C	Semis & Semi Equipment	413	2,209		✓					30.9	9.9	✓
6723	RENESAS ELECTRONIC	Semis & Semi Equipment	387	6,506				✓		✓	13.6	2.6	✓
3436	SUMCO CORPORATION	Semis & Semi Equipment	351	1,091	✓						55.8	1.9	✓
6861	KEYENCE CORP	Tech Hardware & Equipment	349	19,359					✓		32.2	5.6	✓
4063	SHIN-ETSU CHEMICAL	Materials	321	11,690	✓						18.4	2.5	✓
4004	RESONAC HOLDINGS C	Materials	298	2,860	✓		✓			✓	19.6	3.9	✓
7735	SCREEN HOLDINGS CO	Semis & Semi Equipment	244	2,621		✓					19.3	5.3	✓
7012	KAWASAKI HEAVY IND	Capital Goods	239	2,112					✓		15.6	2.3	✗
3110	NITTO BOSEKI CO	Capital Goods	225	562			✓			✓	22.5	3.1	✓
6506	YASKAWA ELEC CORP	Capital Goods	198	1,195					✓		22.1	2.4	✗
7741	HOYA CORP	Health Care Equipment & Svcs	177	7,864	✓						26.9	7.7	✓
6273	SMC CORP	Capital Goods	149	4,447		✓			✓		19.9	2.1	✓
2802	AJINOMOTO CO INC	Food Beverage & Tobacco	143	4,939			✓				31.2	6.3	✗
6526	SOCIONEXT INC	Semis & Semi Equipment	134	404				✓			23.9	3.0	✓
6361	EBARA CORP	Capital Goods	116	2,031		✓				✓	17.8	4.0	✓
6871	MICRONICS JAPAN CO	Semis & Semi Equipment	111	554			✓				18.8	8.1	✓
6787	MEIKO ELECTRONICS	Tech Hardware & Equipment	91	640			✓			✓	18.4	4.7	✓
6315	TOWA CORP	Semis & Semi Equipment	78	165			✓				16.6	2.3	✓
6383	DAIFUKU CO LTD	Capital Goods	74	2,108			✓		✓		20.7	4.5	✗
6890	FERROTEC CORP	Semis & Semi Equipment	69	575	✓	✓				✓	13.3	1.9	✗
5333	NGK CORP	Capital Goods	64	1,549		✓					16.5	1.9	✓
5344	MARUWA CO LTD	Tech Hardware & Equipment	64	688		✓				✓	23.8	4.7	✓
5201	AGC INC	Capital Goods	63	1,242	✓						13.1	0.8	✗
5334	NITERRA CO LTD	Automobiles & Components	62	1,518			✓				12.2	1.9	✗
6590	SHIBAURA MECHATRON	Semis & Semi Equipment	62	264		✓	✓				14.1	4.5	✓
5332	TOTO LTD	Capital Goods	61	984		✓					18.3	1.8	✗
4186	TOKYO OHKA KOGYO	Materials	60	1,068	✓		✓				20.6	4.4	✓
6645	OMRON CORP	Tech Hardware & Equipment	58	1,304					✓		21.0	1.5	✓
6323	RORZE CORP	Semis & Semi Equipment	57	668					✓		18.0	5.0	✓
6227	AIMECHATEC LTD	Semis & Semi Equipment	57	125			✓				22.2	8.6	✓
3407	ASAHI KASEI CORP	Materials	56	2,247			✓				12.4	1.1	✗
7911	TOPPAN HOLDINGS IN	Commercial & Professional Svcs	55	1,449			✓				16.3	1.0	✓
268A	RIGAKU HOLDINGS CO	Tech Hardware & Equipment	52	378		✓					23.6	4.3	✓
6481	THK CO LTD	Capital Goods	49	766					✓		16.4	2.8	✓
7729	TOKYO SEIMITSU CO	Semis & Semi Equipment	48	747			✓				19.4	3.8	✓
4182	MITSUBISHI GAS CHM	Materials	41	827	✓		✓				13.2	1.2	✓
5805	SWCC CORPORATION	Capital Goods	40	417			✓			✓	16.3	4.1	✓
indicate GS covered stocks			Median	37	556						16.9	2.8	

Source: I/B/E/S, FactSet, Goldman Sachs Global Investment Research

Exhibit 67: Full Japan AI semiconductor ecosystem universe (continued)

Data as of Sep 25, 2026



Japan AI semiconductor ecosystem universe

 Japan AI semiconductor ecosystem universe					SM1: Semiconductor materials and consumables								
					SM2: Wafer-fab equipment, metrology & critical subsystem/components						SM3: Advanced packaging, substrates, assembly & test		
					SM4: AI-related semiconductor devices - memory, compute & edge AI								
					SM5: Fab automation, material handling & utility infrastructure								
					Ticker	Name	GICS Industry Group	6M ADTV (\$mn)	Mcap (¥bn)	SM1	SM2	SM3	SM4
6754	ANRITSU CORP	Tech Hardware & Equipment	40	462			✓			✓	23.8	3.3	✓
6855	JAPAN ELECTRON MAT	Semis & Semi Equipment	34	111			✓				14.8	2.5	✓
6856	HORIBA LTD	Tech Hardware & Equipment	33	1,007		✓					16.4	2.9	✓
6728	ULVAC INC	Semis & Semi Equipment	32	376		✓					16.9	1.6	✓
4021	NISSAN CHEMICAL CO	Materials	31	1,124	✓		✓				19.2	4.4	✗
4203	SUMITOMO BAKELITE	Materials	31	722			✓				19.3	2.1	✗
6370	KURITA WATER INDS	Capital Goods	30	897					✓		16.7	2.5	✓
7826	FURUYA METAL CO LT	Tech Hardware & Equipment	30	297	✓						15.0	3.7	✗
6407	CKD CORP	Capital Goods	29	358		✓					15.2	2.3	✓
429A	TEKSCEND PHOTOMASK	Semis & Semi Equipment	29	386	✓						14.1	2.2	✓
6368	ORGANO CORP	Capital Goods	26	588					✓		17.5	4.1	✓
4042	TOSOH CORP	Materials	24	864	✓						12.4	1.0	✗
6622	DAIHEN CORP	Capital Goods	23	322		✓			✓	✓	14.7	2.0	✓
6254	NOMURA MICRO SCIEN	Capital Goods	22	139					✓		10.6	3.3	✓
6941	YAMAICHI ELECTRONI	Semis & Semi Equipment	22	176			✓			✓	13.4	3.4	✓
5301	TOKAI CARBON CO	Materials	22	495	✓						17.2	1.5	✗
4043	TOKUYAMA CORP	Materials	22	304	✓						8.7	1.1	✗
4061	DENKA COMPANY LTD	Materials	21	297			✓				14.0	0.9	✗
7966	LINTEC CORP	Materials	21	417			✓				15.4	1.5	✗
4971	MEC COMPANY LTD	Materials	20	132			✓				18.8	4.0	✗
4022	RASA INDUSTRIES	Materials	19	75	✓						18.3	2.3	✗
4368	FUSO CHEMICAL CO	Materials	18	326	✓						14.9	2.8	✗
6951	JEOL LTD	Tech Hardware & Equipment	16	396		✓					16.1	2.6	✗
6507	SINFONIA TECHNOLOG	Capital Goods	16	378					✓		19.4	3.6	✗
6961	ENPLAS CORP	Tech Hardware & Equipment	16	96			✓				12.8	1.4	✗
4047	KANTO DENKA KOGYO	Materials	15	138	✓						14.4	1.9	✗
5393	NICHIAS CORP	Capital Goods	14	650		✓					17.1	2.7	✗
4369	TRI CHEMICAL LABOR	Semis & Semi Equipment	13	97	✓						15.0	2.7	✗
6925	USHIO INC	Capital Goods	12	335		✓	✓				21.1	1.6	✗
4401	ADEKA CORPORATION	Materials	12	460	✓						14.7	1.5	✗
6055	JAPAN MATERIAL CO	Semis & Semi Equipment	12	243					✓		17.5	3.8	✗
3445	RS TECHNOLOGIES CO	Semis & Semi Equipment	11	163	✓						13.7	2.0	✗
6387	SAMCO INC	Semis & Semi Equipment	10	64		✓					18.4	4.1	✗
6627	TERA PROBE INC	Semis & Semi Equipment	10	109			✓				13.6	2.6	✗
5384	FUJIMI INC	Materials	10	278	✓						19.0	3.1	✗
4626	TAIYO HLDG CO LTD	Materials	10	558			✓				16.1	4.6	✗
4099	SHIKOKU KASEI HOLD	Materials	10	200	✓		✓				15.9	2.1	✗
4966	C. UYEMURA & CO	Materials	9	363			✓				17.3	2.8	✗
6490	PILLAR CORPORATION	Materials	9	259		✓					14.4	3.0	✗
5331	NORITAKE CO LTD	Capital Goods	8	209	✓					✓	12.3	1.2	✗
6266	TAZMO CO LTD	Semis & Semi Equipment	7	56		✓	✓		✓		10.1	2.0	✗
4970	TOYO GOSEI CO LTD	Materials	7	81	✓						15.8	2.9	✗
3449	TECHNOFLEX CORP	Capital Goods	7	94					✓		16.9	3.1	✗
5310	TOYO TANSO CO LTD	Capital Goods	6	159		✓					23.1	1.6	✗
6800	YOKOWO CO LTD	Tech Hardware & Equipment	6	106			✓				15.5	1.7	✗
4187	OSAKA ORGANIC CHEM	Materials	6	95	✓						13.3	1.7	✗
3104	FUJIBO HOLDINGS	Consumer Durables & Apparel	5	113	✓						14.4	2.2	✗
indicate GS covered stocks		Median	37	556							16.9	2.8	

Source: I/B/E/S, FactSet, Goldman Sachs Global Investment Research

Exhibit 68: Full Japan AI data center hardware universe

Data as of Sep 25, 2026

**AI data center hardware**

AI data center hardware					HW1: High-speed interconnect & data transmission						
					HW2: Server, PCB, rack power & cooling components						
					HW3: Data center power & thermal hardware						
Ticker	Name	GICS Industry Group	6M ADTV (\$mn)	Mcap (¥bn)	HW1	HW2	HW3	SM exposure	NTM PE	FY0 PB	In HW basket
5803	FUJIKURA	Capital Goods	1,803	8,851	✓	✓			25.5	14.7	✓
6976	TAIYO YUDEN CO LTD	Technology Hardware & Equipment	1,599	1,260		✓			30.1	3.3	✓
6981	MURATA MFG CO	Technology Hardware & Equipment	1,373	15,747		✓			33.4	5.4	✓
5801	FURUKAWA ELECTRIC	Capital Goods	1,010	2,704	✓	✓			21.7	6.5	✓
5016	JX ADVANCED METALS	Materials	671	3,374	✓	✓		✓	18.4	4.5	✓
5706	MITSUI KINZOKU CO	Materials	448	1,319	✓	✓		✓	13.7	3.2	✓
5802	SUMITOMO ELECTRIC	Automobiles & Components	406	7,022	✓				17.8	2.5	✓
6501	HITACHI	Capital Goods	403	25,109			✓		23.0	3.8	✓
6723	RENESAS ELECTRONIC	Semiconductors & Semiconductor Equipment	387	6,506		✓		✓	13.6	2.6	✓
4004	RESONAC HOLDINGS C	Materials	298	2,860		✓		✓	19.6	3.9	✓
6762	TDK CORP	Technology Hardware & Equipment	272	5,789		✓			20.3	2.6	✓
6752	PANASONIC HLDGS CO	Consumer Durables & Apparel	269	11,262		✓	✓		19.9	2.1	✓
6503	MITSUBISHI ELEC CP	Capital Goods	232	11,071	✓		✓		18.9	2.4	✓
3110	NITTO BOSEKI CO	Capital Goods	225	562		✓		✓	22.5	3.1	✓
6963	ROHM CO LTD	Semiconductors & Semiconductor Equipment	146	1,935		✓			36.3	2.4	✗
6361	EBARA CORP	Capital Goods	116	2,031			✓	✓	17.8	4.0	✓
6594	NIDEC CORPORATION	Capital Goods	96	3,387		✓	✓		16.4	1.9	✓
6787	MEIKO ELECTRONICS	Technology Hardware & Equipment	91	640		✓		✓	18.4	4.7	✓
6504	FUJI ELECTRIC CO.	Capital Goods	88	2,033			✓		16.9	2.5	✗
6479	MINEBEA MITSUMI	Capital Goods	77	1,535		✓			15.2	1.6	✓
6890	FERROTEC CORP	Semiconductors & Semiconductor Equipment	69	575	✓			✓	13.3	1.9	✗
5344	MARUWA CO LTD	Technology Hardware & Equipment	64	688		✓		✓	23.8	4.7	✓
6965	HAMAMATSU PHOTONIC	Technology Hardware & Equipment	58	748	✓				33.3	2.2	✓
4980	DEXERIALS CORP	Technology Hardware & Equipment	49	538	✓				16.5	4.7	✓
6834	SEIKOH GIKEN CO	Technology Hardware & Equipment	43	284	✓				26.4	8.0	✓
5805	SWCC CORPORATION	Capital Goods	40	417	✓			✓	16.3	4.1	✓
6754	ANRITSU CORP	Technology Hardware & Equipment	40	462	✓			✓	23.8	3.3	✓
6134	FUJI CORPORATION	Capital Goods	39	729		✓			13.4	2.8	✓
6278	UNION TOOL CO	Capital Goods	38	283		✓			18.1	3.1	✓
6674	GS YUASA CORP	Automobiles & Components	36	542			✓		12.5	1.4	✗
6779	NIHON DEMP KOGYO	Technology Hardware & Equipment	31	75	✓				21.7	2.3	✓
6806	HIROSE ELECTRIC	Technology Hardware & Equipment	29	864	✓				19.4	2.2	✗
6997	NIPPON CHEMI-CON	Technology Hardware & Equipment	29	73		✓			12.7	1.3	✓
6777	SANTEC HLDGS CORP	Technology Hardware & Equipment	29	262	✓				23.9	9.2	✓
6996	NICHICON CORP	Technology Hardware & Equipment	25	183		✓			18.3	1.5	✗
6622	DAIHEN CORP	Capital Goods	23	322			✓	✓	14.7	2.0	✓
6941	YAMAICHI ELECTRONI	Semiconductors & Semiconductor Equipment	22	176	✓			✓	13.4	3.4	✓
6508	MEIDENSHA CORP	Capital Goods	17	522			✓		18.6	3.0	✗
4461	DKS CO. LTD.	Materials	13	149		✓			14.2	2.8	✗
6999	KOA CORPORATION	Technology Hardware & Equipment	11	86		✓			12.8	1.0	✗
3891	NIPPON KODOSHI	Technology Hardware & Equipment	10	73		✓			14.1	2.8	✗
6707	SANKEN ELECTRIC CO	Semiconductors & Semiconductor Equipment	10	144		✓	✓		61.3	1.1	✗
5331	NORITAKE CO LTD	Capital Goods	8	209		✓		✓	12.3	1.2	✗
6617	TAKAOKA TOKO CO LT	Capital Goods	7	113			✓		9.8	1.6	✗
6516	SANYO DENKI CO	Capital Goods	6	211		✓	✓		13.9	1.5	✗
	indicate GS covered stocks	Median	58	640					18.3	2.8	

Source: I/B/E/S, FactSet, Goldman Sachs Global Investment Research

Exhibit 69: Exposure to AI based on GS Asia Tech report and RBICS sales exposure data

Data as of Sep 25, 2026

Ticker	Name	GICS Industry Group	AISM	AIHW	6M ADTV (\$ mn)	Mcap (\$ bn)	Exposure based on Asia Tech Report				RBICS select sub-industries sales exposure (%)
							Sales AI exposure FY25 (%)	FY28E (%)	OP AI exposure FY25 (%)	FY28E (%)	
285A	KIOXIA HLDGS CORP	Semis & Semi Equipment	✓		14,248	30,603	45	50	na	na	87
8035	TOKYO ELECTRON	Semis & Semi Equipment	✓		1,412	26,453	40	45	na	na	100
6501	HITACHI	Capital Goods		✓	403	25,109	13	19	15	31	40
6857	ADVANTEST CORP	Semis & Semi Equipment	✓		1,742	24,888	65	70	na	na	100
6861	KEYENCE CORP	Tech Hardware & Equipment	✓		349	19,359	na	na	na	na	100
6981	MURATA MFG CO	Tech Hardware & Equipment		✓	1,373	15,747	10	39	12	44	93
4063	SHIN-ETSU CHEMICAL	Materials	✓		321	11,690	13	24	18	32	62
6752	PANASONIC HLDGS CO	Consumer Durables & Apparel		✓	269	11,262	7	16	27	47	29
6503	mitsubishi elec CP	Capital Goods		✓	232	11,071	4	8	12	20	59
5803	FUJIKURA	Capital Goods		✓	1,803	8,851	39	55	69	90	80
7741	HOYA CORP	Health Care Equipment & Services	✓		177	7,864	20	25	na	na	37
5802	SUMITOMO ELECTRIC	Automobiles & Components		✓	406	7,022	3	12	15	42	35
4062	IBIDEN CO LTD	Tech Hardware & Equipment	✓		1,013	6,621	46	72	74	89	78
6723	RENESAS ELECTRONIC	Semis & Semi Equipment	✓	✓	387	6,506	12	25	12	25	52
6146	DISCO CORPORATION	Semis & Semi Equipment	✓		772	5,884	65	70	na	na	100
6762	TDK CORP	Tech Hardware & Equipment		✓	272	5,789	10	22	12	25	45
2802	AJINOMOTO CO INC	Food Beverage & Tobacco	✓		143	4,939	na	na	na	na	6
6273	SMC CORP	Capital Goods	✓		149	4,447	na	na	na	na	100
6920	LASERTEC CORP	Semis & Semi Equipment	✓		947	3,857	55	60	na	na	100
6594	NIDEC CORPORATION	Capital Goods		✓	96	3,387	17	21	na	30	19
5016	JX ADVANCED METALS	Materials	✓	✓	671	3,374	9	24	12	32	20
4004	RESONAC HOLDINGS C	Materials	✓	✓	298	2,860	10	28	31	71	44
5801	FURUKAWA ELECTRIC	Capital Goods		✓	1,010	2,704	12	33	41	72	39
7735	SCREEN HOLDINGS CO	Semis & Semi Equipment	✓		244	2,621	20	30	na	na	92
3407	ASAHI KASEI CORP	Materials	✓		56	2,247	2	4	5	13	42
6525	KOKUSAI ELECTRIC C	Semis & Semi Equipment	✓		413	2,209	40	40	na	na	100
7012	KAWASAKI HEAVY IND	Capital Goods	✓		239	2,112	na	na	na	na	15
6383	DAIFUKU CO LTD	Capital Goods	✓		74	2,108	na	na	na	na	38
6504	FUJI ELECTRIC CO.	Capital Goods		✓	88	2,033	4	8	9	19	50
6361	EBARA CORP	Capital Goods	✓	✓	116	2,031	15	25	na	na	90
6963	ROHM CO LTD	Semis & Semi Equipment		✓	146	1,935	5	10	5	13	100
5333	NGK CORP	Capital Goods	✓		64	1,549	22	30	34	39	35
6479	MINEBEA MITSUMI	Capital Goods		✓	77	1,535	6	9	16	22	63
5334	NITERRA CO LTD	Automobiles & Components	✓		62	1,518	8	10	8	13	18
7911	TOPPAN HOLDINGS IN	Commercial & Professional Services	✓		55	1,449	5	5	15	25	10
5706	mitsui KINZOKU CO	Materials	✓	✓	448	1,319	7	22	12	34	42
6645	OMRON CORP	Tech Hardware & Equipment	✓		58	1,304	na	na	na	na	53
6976	TAIYO YUDEN CO LTD	Tech Hardware & Equipment		✓	1,599	1,260	6	27	8	40	93
5201	AGC INC	Capital Goods	✓		63	1,242	na	na	na	na	10
6506	YASKAWA ELEC CORP	Capital Goods	✓		198	1,195	na	na	na	na	44
4021	NISSAN CHEMICAL CO	Materials	✓		31	1,124	na	na	na	na	37
3436	SUMCO CORPORATION	Semis & Semi Equipment	✓		351	1,091	14	43	46	53	100
4186	TOKYO OHKA KOGYO	Materials	✓		60	1,068	18	43	24	57	100
6856	HORIBA LTD	Tech Hardware & Equipment	✓		33	1,007	na	na	na	na	87
5332	TOTO LTD	Capital Goods	✓		61	984	na	na	na	na	9
6370	KURITA WATER INDS	Capital Goods	✓		30	897	na	na	na	na	63
6806	HIROSE ELECTRIC	Tech Hardware & Equipment		✓	29	864	5	12	5	15	100
4042	TOSOH CORP	Materials	✓		24	864	na	na	na	na	37
4182	mitsubishi GAS CHM	Materials	✓		41	827	4	16	16	46	38
6481	THK CO LTD	Capital Goods	✓		49	766	na	na	na	na	100
6965	HAMAMATSU PHOTONIC	Tech Hardware & Equipment		✓	58	748	na	na	na	na	100
7729	TOKYO SEIMITSU CO	Semis & Semi Equipment	✓		48	747	30	40	na	na	100
6134	FUJI CORPORATION	Capital Goods		✓	39	729	na	na	na	na	90
4203	SUMITOMO BAKELITE	Materials	✓		31	722	na	na	na	na	33
5344	MARUWA CO LTD	Tech Hardware & Equipment	✓	✓	64	688	20	40	25	45	86
6323	RORZE CORP	Semis & Semi Equipment	✓		57	668	na	na	na	na	94
5393	NICHIAS CORP	Capital Goods	✓		14	650	na	na	na	na	16
6787	MEIKO ELECTRONICS	Tech Hardware & Equipment		✓	91	640	na	na	na	na	100
6368	ORGANO CORP	Capital Goods	✓		26	588	na	na	na	na	86
6890	FERROTEC CORP	Semis & Semi Equipment	✓	✓	69	575	na	na	na	na	40
3110	NITTO BOSEKI CO	Capital Goods	✓	✓	225	562	28	49	66	88	42
4626	TAIYO HLDG CO LTD	Materials	✓		10	558	na	na	na	na	69

indicate GS covered stocks

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 70: Exposure to AI based on GS Asia Tech report and RBICS sales exposure data

Data as of Sep 25, 2026

Ticker	Name	GICS Industry Group	AISM	AIHW	6M ADTV (\$ mn)	Mcap (¥ bn)	Exposure based on Asia Tech Report				RBICS select
							Sales AI exposure FY25 (%)	FY28E (%)	OP AI exposure FY25 (%)	FY28E (%)	sub-industries sales exposure (%)
6871	MICRONICS JAPAN CO	Semis & Semi Equipment	✓		111	554	na	na	na	na	100
6674	GS YUASA CORP	Automobiles & Components		✓	36	542	na	na	na	na	15
4980	DEXERIALS CORP	Tech Hardware & Equipment		✓	49	538	na	na	na	na	100
6508	MEIDENSHA CORP	Capital Goods		✓	17	522	6	8	10	16	62
5301	TOKAI CARBON CO	Materials	✓		22	495	na	na	na	na	48
6754	ANRITSU CORP	Tech Hardware & Equipment	✓	✓	40	462	19	32	28	55	100
4401	ADEKA CORPORATION	Materials	✓		12	460	na	na	na	na	52
5805	SWCC CORPORATION	Capital Goods	✓	✓	40	417	9	15	14	34	97
7966	LINTEC CORP	Materials	✓		21	417	na	na	na	na	29
6526	SOCIONEXT INC	Semis & Semi Equipment	✓		134	404	na	na	na	na	100
6951	JEOL LTD	Tech Hardware & Equipment	✓		16	396	10	20	na	na	27
429A	TEKSCEND PHOTOMASK	Semis & Semi Equipment	✓		29	386	na	na	na	na	100
6507	SINFONIA TECHNOLOG	Capital Goods	✓		16	378	na	na	na	na	22
268A	RIGAKU HOLDINGS CO	Tech Hardware & Equipment	✓		52	378	na	na	na	na	100
6728	ULVAC INC	Semis & Semi Equipment	✓		32	376	10	20	na	na	56
4966	C. UYEMURA & CO	Materials	✓		9	363	na	na	na	na	99
6407	CKD CORP	Capital Goods	✓		29	358	na	na	na	na	96
6925	USHIO INC	Capital Goods	✓		12	335	na	na	na	na	50
4368	FUSO CHEMICAL CO	Materials	✓		18	326	na	na	na	na	54
6622	DAIHEN CORP	Capital Goods	✓	✓	23	322	14	20	25	39	54
4043	TOKUYAMA CORP	Materials	✓		22	304	na	na	na	na	26
7826	FURUYA METAL CO LT	Tech Hardware & Equipment	✓		30	297	na	na	na	na	28
4061	DENKA COMPANY LTD	Materials	✓		21	297	2	6	6	18	25
6834	SEIKOH GIKEN CO	Tech Hardware & Equipment		✓	43	284	na	na	na	na	53
6278	UNION TOOL CO	Capital Goods		✓	38	283	na	na	na	na	100
5384	FUJIMI INC	Materials	✓		10	278	21	40	26	44	100
6590	SHIBAURA MECHATRON	Semis & Semi Equipment	✓		62	264	na	na	na	na	95
6777	SANTEC HLDGS CORP	Tech Hardware & Equipment		✓	29	262	na	na	na	na	91
6490	PILLAR CORPORATION	Materials	✓		9	259	na	na	na	na	66
6055	JAPAN MATERIAL CO	Semis & Semi Equipment	✓		12	243	na	na	na	na	69
6516	SANYO DENKI CO	Capital Goods		✓	6	211	na	na	na	na	100
5331	NORITAKE CO LTD	Capital Goods	✓	✓	8	209	na	na	na	na	56
4099	SHIKOKU KASEI HOLD	Materials	✓		10	200	na	na	na	na	26
6996	NICHICON CORP	Tech Hardware & Equipment	✓	✓	25	183	3	13	8	17	100
6941	YAMAICHI ELECTRONI	Semis & Semi Equipment	✓	✓	22	176	na	na	na	na	100
6315	TOWA CORP	Semis & Semi Equipment	✓		78	165	na	na	na	na	95
3445	RS TECHNOLOGIES CO	Semis & Semi Equipment	✓		11	163	na	na	na	na	100
5310	TOYO TANSO CO LTD	Capital Goods	✓		6	159	na	na	na	na	67
4461	DKS CO. LTD.	Materials		✓	13	149	na	na	na	na	55
6707	SANKEN ELECTRIC CO	Semis & Semi Equipment		✓	10	144	na	na	na	na	100
6254	NOMURA MICRO SCIEN	Capital Goods	✓		22	139	na	na	na	na	100
4047	KANTO DENKA KOGYO	Materials	✓		15	138	na	na	na	na	84
4971	MEC COMPANY LTD	Materials	✓		20	132	na	na	na	na	99
6227	AIMECHATEC LTD	Semis & Semi Equipment	✓		57	125	na	na	na	na	93
6617	TAKAOKA TOKO CO LT	Capital Goods		✓	7	113	na	na	na	na	88
3104	FUJIBO HOLDINGS	Consumer Durables & Apparel	✓		5	113	na	na	na	na	37
6855	JAPAN ELECTRON MAT	Semis & Semi Equipment	✓		34	111	na	na	na	na	100
6627	TERA PROBE INC	Semis & Semi Equipment	✓		10	109	na	na	na	na	100
6800	YOKOWO CO LTD	Tech Hardware & Equipment	✓		6	106	na	na	na	na	34
4369	TRI CHEMICAL LABOR	Semis & Semi Equipment	✓		13	97	15	34	17	37	100
6961	ENPLAS CORP	Tech Hardware & Equipment	✓		16	96	na	na	na	na	55
4187	OSAKA ORGANIC CHEM	Materials	✓		6	95	na	na	na	na	46
3449	TECHNOFLEX CORP	Capital Goods	✓		7	94	na	na	na	na	93
6999	KOA CORPORATION	Tech Hardware & Equipment		✓	11	86	na	na	na	na	100
4970	TOYO GOSEI CO LTD	Materials	✓		7	81	na	na	na	na	100
4022	RASA INDUSTRIES	Materials	✓		19	75	na	na	na	na	89
6779	NIHON DEMPA KOGYO	Tech Hardware & Equipment		✓	31	75	na	na	na	na	100
3891	NIPPON KODOSHI	Tech Hardware & Equipment		✓	10	73	na	na	na	na	100
6997	NIPPON CHEMI-CON	Tech Hardware & Equipment		✓	29	73	na	na	na	na	100
6387	SAMCO INC	Semis & Semi Equipment	✓		10	64	na	na	na	na	100
6266	TAZMO CO LTD	Semis & Semi Equipment	✓		7	56	na	na	na	na	97

indicate GS covered stocks

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 71: GICS industries excluded in the AI SM/HW universe

GICS Sector	GICS Industry Group	GICS Industry
Communication Services	Media & Entertainment	Entertainment
Communication Services	Media & Entertainment	Interactive Media & Services
Communication Services	Media & Entertainment	Media
Communication Services	Telecommunication Services	Diversified Telecommunication Services
Communication Services	Telecommunication Services	Wireless Telecommunication Services
Consumer Discretionary	Automobiles & Components	Automobiles
Consumer Discretionary	Consumer Discretionary Distribution & Retail	Broadline Retail
Consumer Discretionary	Consumer Discretionary Distribution & Retail	Distributors
Consumer Discretionary	Consumer Discretionary Distribution & Retail	Specialty Retail
Consumer Discretionary	Consumer Durables & Apparel	Leisure Products
Consumer Discretionary	Consumer Services	Diversified Consumer Services
Consumer Discretionary	Consumer Services	Hotels Restaurants & Leisure
Consumer Staples	Consumer Staples Distribution & Retail	Consumer Staples Distribution & Retail
Consumer Staples	Food Beverage & Tobacco	Beverages
Consumer Staples	Food Beverage & Tobacco	Tobacco
Consumer Staples	Household & Personal Products	Household Products
Consumer Staples	Household & Personal Products	Personal Care Products
Energy	Energy	Energy Equipment & Services
Energy	Energy	Oil Gas & Consumable Fuels
Financials	Banks	Banks
Financials	Financial Services	Capital Markets
Financials	Financial Services	Consumer Finance
Financials	Financial Services	Financial Services
Financials	Insurance	Insurance
Health Care	Health Care Equipment & Services	Health Care Providers & Services
Health Care	Health Care Equipment & Services	Health Care Technology
Health Care	Pharmaceuticals Biotechnology & Life Sciences	Biotechnology
Health Care	Pharmaceuticals Biotechnology & Life Sciences	Life Sciences Tools & Services
Health Care	Pharmaceuticals Biotechnology & Life Sciences	Pharmaceuticals
Industrials	Capital Goods	Aerospace & Defense
Industrials	Capital Goods	Construction & Engineering
Industrials	Capital Goods	Trading Companies & Distributors
Industrials	Commercial & Professional Services	Professional Services
Industrials	Transportation	Air Freight & Logistics
Industrials	Transportation	Ground Transportation
Industrials	Transportation	Marine Transportation
Industrials	Transportation	Passenger Airlines
Industrials	Transportation	Transportation Infrastructure
Information Technology	Software & Services	IT Services
Information Technology	Software & Services	Software
Materials	Materials	Construction Materials
Materials	Materials	Containers & Packaging
Materials	Materials	Paper & Forest Products
Real Estate	Real Estate Management & Development	Real Estate Management & Development
Utilities	Utilities	Electric Utilities
Utilities	Utilities	Gas Utilities
Utilities	Utilities	Independent Power and Renewable Electricity Producers

Source: FactSet

Exhibit 72: Stock universe is first filtered based on exposure to any RBICS sub-industries with potential AI exposure

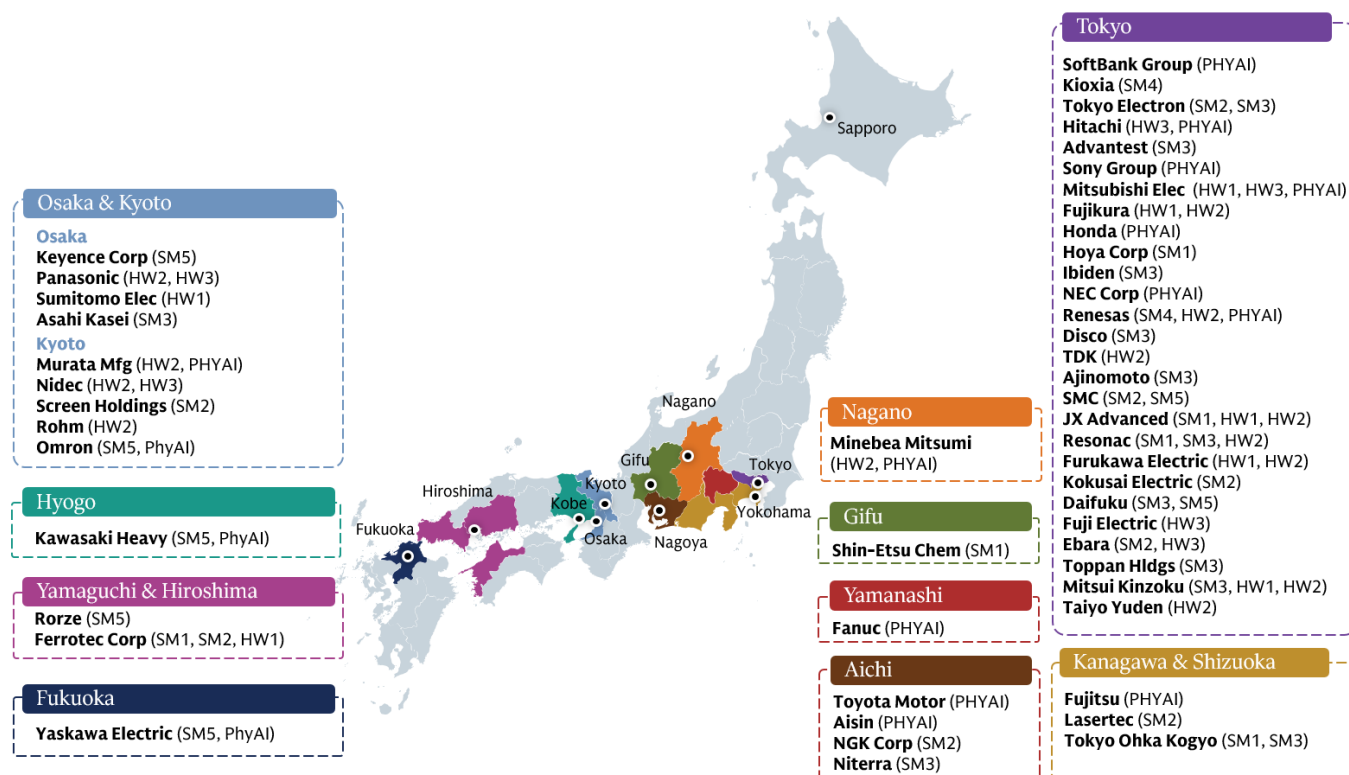
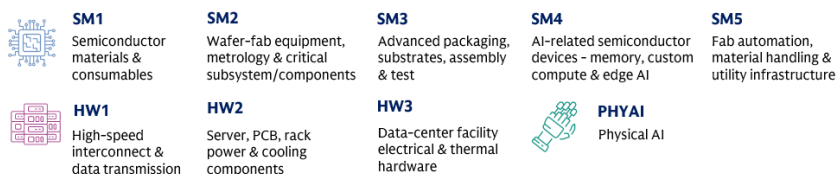
RBICS sub-industries identified as potentially AI SM/HW-related	
Air, Liquid and Gas Control Equipment Products	Motion Control and Precision Motors Manufacturing
Assembly Equipment Manufacturing	Optoelectronics Electronic Components
Backup, Emergency and Standby Power Products	Oscillator and Piezoelectric Passive Components
Cable Interconnect Components	Other Discrete Semiconductors
Carbon and Graphite Products Manufacturing	Other Front End Processing Equipment Makers
Clay and Ceramic Products Manufacturing	Other Interconnect Components
Compressor and Pumping Equipment Manufacturing	Other Optoelectronics Discrete Semiconductors
Data Storage Drives and Peripherals	Other Organic Chemical Manufacturers
Diversified Cleaning and Sanitizing Agents Makers	Other Passive Capacitor Electronic Components
Diversified Electrical/Power System Manufacturing	Other Passive Electronic Components
Diversified Industrial Manufacturing	Other Performance Additive Manufacturing
Diversified Semiconductor Capital Equipment Makers	Other Power Analog and Mixed Signal Semiconductors
Diversified Semiconductors	Other Programmable Logic and ASIC Semiconductors
Diversified Specialty/Performance Chemicals Makers	Other Test and Measurement Equipment
Electronic Interconnect Components	Other Wide Area Networking (WAN) Equipment
Electronic Materials Manufacturing	Packaging and Coating Application Machinery Makers
Electronics Manufacturing Equipment Manufacturing	Photolithography Equipment Manufacturing
Fabricated Pipes and Pipe Fittings Products	Pipes and Pipe Fittings Manufacturing
Flash Memory Semiconductors	Polymer Additives Manufacturing
General Communications Equipment	Power Generation/Support Products Manufacturing
General Factory Automation Makers	Power Module and Subassembly Electronic Components
General Plastic and Rubber Manufacturing	Power Transmission and Distribution Products
General Purpose Test and Measurement Equipment	Power, Control and Mixed Signal Semiconductors
Heating, Ventilation and Air Conditioning Products	Printed Circuit Board/Wiring Board Manufacturing
Lasers and Optical Instrument Manufacturing	Printed Circuit Boards
Lighting Equipment and Components Manufacturing	Semiconductor Capital Equipment/Parts Distribution
Liquid and Water Purification/Filtration Products	Semiconductor Components/Subsystems Manufacturing
Machine Vision and Quality Control Manufacturing	Semiconductor Testing Services
Magnetic Passive Electronic Components	Server Computer Systems
Mechanical Power Transmission (MPT) Manufacturing	Specialty Analog and Mixed Signal Semiconductors
Microprocessor (MPU) Semiconductors	Test, Measurement and Metrology Equipment Makers
Mixed Heavy-Duty and High-End Batteries Makers	Valves and Fluid Control Products
Mixed Module and Subassembly Electronic Components	Wafer Blank Makers and Equipment Manufacturing
Mixed Specialty and Commodity Chemical Makers	Wafer Processing Subsystem Equipment Manufacturing
Monitoring and Control Sensor/Instrument Products	Wires and Cables Manufacturing

Source: FactSet

We provide a map showing the headquarters of notable companies in our Japan AI universe. For clients coming to Japan to meet with corporate management to discuss specific layers of the AI value chain we hope that this map can be a useful logistical tool.

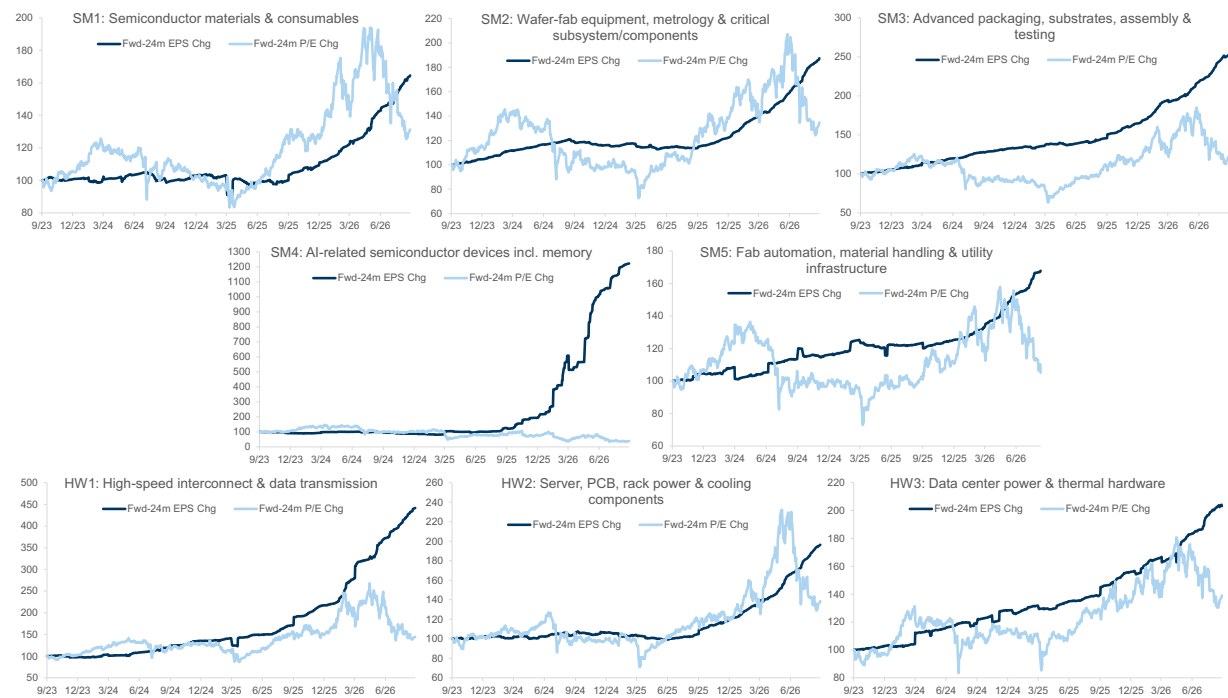
Exhibit 73: Japan AI infrastructure breakdown by region

Japan AI Infrastructure Ecosystem Map



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 74: Performance breakdown of each AI SM/HW layers
Data as of Sep 25, 2026



Rebalanced semi-annually based on average of liquidity and market cap

Source: FactSet, Goldman Sachs

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